

A Structure-Aware Hierarchical Graph Neural Network for Multi-Label Scent Profile Prediction in Binary Fragrance Mixtures

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Department of Computer Science
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We declare that the Senior Project entitled

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which we submitted to the faculty of the

Department of Computer Science, Ateneo de Naga University

is our own work. To the best of our knowledge, it does not contain materials published or written by another person, except where due citation and acknowledgement is made in our senior project documentation. The contributions of other people whom we worked with to complete this senior project are explicitly cited and acknowledged in our senior project documentation.

We also declare that the intellectual content of this senior project is the product of our own work. We conceptualized, designed, encoded, and debugged the source code of the core programs in our senior project. The source code of third party APIs and library functions used in our program are explicitly cited and acknowledged in our senior project documentation. Also duly acknowledged are the assistance of others in minor details of editing and reproduction of the documentation.

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ABSTRACT

Predicting the scent of a blend of two aroma chemicals is harder than predicting the scent of one, because a blend does not always smell like the sum of its parts. Graph neural networks (GNNs) now predict the scent of single molecules well, but when extended to blends they usually combine the two molecules by averaging, which assumes the molecules do not affect each other. This study tests whether adding an interaction step improves prediction. Each molecule is encoded by a shared graph attention network, and the two resulting representations are combined in one of three ways: by averaging, by concatenation (a reproduction of a published pairwise model), or by a Set Transformer that adjusts each representation according to its partner before combining them. All three models predict a set of scent descriptors for each blend. The models are trained and tested on a public dataset of two-molecule blends whose labels come from the GoodScents reference catalog. Since these labels are not direct human panel measurements, this study first measures their reliability through agreement between independent catalog sources and label statistics. Each blend descriptor is graded into one of three reliability tiers: Gold, where the blend pairing is independently confirmed by both molecules and the descriptor shows reliable cross-catalog agreement between GoodScents and Leffingwell; Silver, where only one of these two conditions holds; and Bronze, where neither holds. Within a leakage-resistant test split built from UMAP and K-means clustering, the models are compared on the high-confidence (Gold) subset using macro F1, micro F1, PR-AUC, and Hamming loss, with paired bootstrap confidence intervals. Finally, the models’ sensitivity to injected label noise is evaluated, to test whether the interaction-aware model degrades faster or slower than the baselines as label quality declines.

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Chapter 1

Introduction

Perfumers and flavorists often use multiple aroma chemicals and combine them to produce their products [37]. Predicting how combinations of aroma chemical ingredients is difficult, since a blend does not always perform additively, or smell like the sum of its parts [43]. A note in one ingredient may be present but disappear in the blend, and occasionally a note may emerge that neither of ingredients has its own [43, 47]. Existing machine learning models, particularly Graph Neural Networks (GNNs), have performed well on predicting the scent of a single molecule from its structure with accuracy comparable to trained human panelists [27]. However, a limitation of these models is that they were not designed for two molecules at once. When models are extended to blends, the usual approach is to implement naive methods such as mean or sum aggregation, which assumes that the two molecules does not affect each other [47]. Recent human-panel evidence suggests that this assumption often holds, while catalog-labeled blend data suggests it often does not [34]. This study examines whether a model that lets the two molecule summaries interact [55] before combining them predicts the scent labels of two-molecule blends better than one that simply averages them [28].

A further complication is that the blend labels used to train and evaluate such models are not direct measurements of human perception. Public blend datasets, such as the one built by Sisson et al. [43], are compiled from reference-catalog recommendations rather than a trained sensory panel, and their reliability has not been independently measured. This study therefore treats label reliability as a first-order question: before comparing how well an interaction-aware

model predicts blend scent, the reliability of the labels themselves is measured through agreement between independent catalog sources, and the resulting comparisons are reported on a graded, high-confidence subset of the data.

1.1 Project Context

Fragrance communities and scent repositories such as Basenotes and Fragrantica have become a common source of olfactory descriptions. The narratives and scent labels, however, are inconsistent across brands and consumers that create confusion between descriptors that smell alike [7, 52]. Reference catalogs such as The Goodscents Company, and Leffingwell & Associates have been used to train state-of-the-art models [28, 43]. These catalogs are more consistent than community reviews, but their descriptors were not all produced by a controlled sensory panel. This study therefore measures the reliability of the labels it uses before relying on them.

Computational olfaction addresses part of this problem through Geometric Deep Learning by adopting modeling structure-odor relationships (SOR) directly from the nominal chemical structures of chemical compounds rather than relying from written scent descriptions. Graph Neural Networks (GNN) frameworks have become the leading approach of this task, using localized message-passing to learn representations of molecular graphs [33, 43]. A structure to perception study has successfully modeled GNN on expertly curated single-molecule datasets, mapping molecular structure to human perception. The study demonstrated that their model’s prediction of a new molecule is closer to the mean of the 15-member human panel [27]. However, while single-molecule GNNs excel at single-graph classification, single-molecule GNNs are not designed to model a multi-graph set representation. This is the reason why it limits the capability to capture the perceptual and physical dynamics of multi-molecule mixtures, such as masking and emergence [49].

Scents are typically blends with multiple components rather than single compounds. When two aroma chemicals get mixed, the result isn’t just additive; it triggers complex psychophysical and physical interactions. In a study conducted by Sisson et al., they demonstrated that in an unweighted binary mixture, blends exhibited psychophysical distortions, losing an average of 1.2 constituent notes to suppression and gaining 0.2 novel descriptors through configural emergence [43]. Current machine learning models struggle to implement a solution for capturing these biological distortions, largely because they rely on naive linear aggregation, such as mean or sum pooling [57]. Averaging

two molecule embeddings can make structurally different pairs indistinguishable to the model: just as the pairs (2, 8) and (5, 5) share the same average, two chemically distinct blends can average to the same combined representation, discarding exactly the information needed to predict suppression or emergence [47].

1.2 Purpose and Description

This study builds a computational model that predicts blend-level odor descriptors directly from molecular structure, and evaluates that model against a reliability-graded version of its own training data, rather than assuming the underlying labels are trustworthy. This study focuses on predicting the emergent, blend-level odor descriptors of binary (two-molecule) mixtures using hierarchical Graph Neural Networks (GNN) with Set Transformers [28, 43]. Three ways of combining the two molecule representations will be compared: mean pooling, concatenation, and our own model which uses Set Transformer. This study first grades every blend label into a reliability tier and reports how the comparison holds up as label confidence changes.

The level 1 consists of the Intra-Molecular GAT Encoder. A Graph Attention Network (GAT) encodes individual aroma-chemical topologies, producing a molecular representation for each molecule based on its own node attributes and covalent bond features [55]. The encoder uses deep message-passing layers to approximate the mapping from a molecular graph to its isolated, elemental scent descriptors, without yet considering how two molecules will interact.

The level 2 serves as the Inter-Molecular Set Aggregator. This extends the single-molecule GNN architecture into our SA-HGNN (Structure-Aware Hierarchical Graph Neural Network) framework using permutation-invariant mechanisms [28, 57]. These enhancements consist of Self-Attention Blocks (SAB) and Pooling by Multihead Attention (PMA). Within this layer, the PMA module combines the pair of molecule embeddings produced by the encoder into a single, learned blend embedding shaped by interaction between the two molecules, rather than by their individual properties alone. Whether this interactive embedding meaningfully outperforms the averaging and concatenation baselines is evaluated specifically on the high-confidence (Gold-tier) subset of blends, so that any observed advantage cannot be attributed to noise in the underlying labels.

The Multi-Label Classification Head acts as the final output layer. A dense feedforward network maps the combined molecule representation, produced by whichever of the three combination

methods is being evaluated, to independent sigmoid activations over the target scent descriptors. All three configurations are evaluated under the same reliability-tiered protocol, so that differences in reported performance reflect the combination method itself rather than uneven exposure to unreliable labels.

The scent classification model is intended to benefit the following entities:

- **Fragrance Chemists and Perfumers.** A structure-grounded screening tool that models how chemicals interact in a blend can help reduce the cost of trial-and-error cycles during perfume formulation [37].
- **Green Chemical Engineers.** This study contributes a standardized comparison of combination methods for two-molecule mixture prediction, evaluated not only for overall accuracy but for how each method’s performance holds up as label reliability declines. It provides empirical evidence on whether attention-based set decoders overcome the “symmetry collapse” limitation of linear graph readouts [47], and on how set-structured models trained on catalog-derived, noisily-labeled data respond to varying label quality [32].
- **Researchers and Maintainers of Olfactory Reference Catalogs.** The reciprocal-agreement and cross-catalog reliability audit developed in this study is not specific to any one dataset. It offers a reusable procedure for quantifying label trustworthiness in future catalog-derived olfactory datasets, including the very catalogs (GoodScents, Leffingwell) this study draws on, without requiring new human sensory panels to do so.

Research Questions

This study seeks to answer the following questions:

1. How consistent are the two-molecule blend labels, based on (a) cross-catalog reciprocal agreement on each pair and (b) GoodScents-Leffingwell agreement on the parent molecules’ descriptors, and how many blends and descriptors survive when only consistent labels are kept?
2. How does the interaction model compare to the averaging and concatenation baselines on the high-confidence test set, and does this comparison change when training on the full dataset versus high-confidence blends only?

3. How does each model’s performance degrade under increasing label noise, and does the interaction model degrade faster or slower than the baselines?

1.3 Objectives

General Objective

To construct a reliability-graded version of a public two-molecule blend dataset and determine whether a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN), pairing a single-molecule Graph Attention Network (GAT) with a Set Transformer interaction layer, predicts multi-label scent profiles more accurately than averaging and concatenation baselines, and whether that conclusion holds as label reliability varies.

To achieve the general objective, the following specific objectives must be accomplished:

- To preprocess the single-molecule datasets from GoodScents and Leffingwell, and pair them with the binary mixture scent records.
- To measure blend label reliability through reciprocal agreement between catalog records of the same pair and cross-catalog agreement between GoodScents and Leffingwell on the parent molecules’ descriptors, and to assign each blend descriptor a reliability tier (Gold, Silver, or Bronze) on that basis.
- To build the Level 1 Graph Attention Network (GAT) that translates individual chemical structures, specifically their atoms and bonds, into uniform scent vectors, and to pre-train it on individual compounds so the model learns basic structure-to-odor relationships before evaluation on mixtures.
- To implement three ways of combining the two molecule embeddings produced by the Level 1 encoder: averaging, concatenation, and a Level 2 Set Transformer interaction layer using Self-Attention Blocks (SAB) and Pooling by Multihead Attention (PMA), so that the interaction model can be compared directly against the two baselines.
- To implement a Multi-Label Classification Head that maps each combined molecule representation to independent sigmoid activations, predicting the presence of up to 74 catalog-recorded scent notes, with the final count determined by the survival filter after graph carving.

- To partition the dataset into training and test sets using UMAP clustering on molecular fingerprints, so that structurally similar molecules are kept together and no test molecule’s family appears in training.
- To compare the three combination methods on the high-confidence (Gold-tier) test partition, training each on (i) the full dataset and (ii) Gold-tier blends only, using macro F1, micro F1, PR-AUC, and Hamming loss with paired bootstrap confidence intervals.
- To evaluate each model’s sensitivity to label noise by retraining all three combination methods on the Gold-tier training partition with controlled noise injected at 0%, 10%, 20%, and 30%, and comparing the resulting decline in macro F1 across models.

1.4 Scope and Limitations

This study focuses on the design, development, and evaluation of a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN), restricted to pairwise, two-molecule blends. The framework uses a two-level architecture to model blending dynamics directly, rather than relying on simple molecular averaging or naive pooling [43, 47]. The first layer is a Level 1 Intra-Molecular Graph Attention Network (GAT) encoder that processes 2D molecular graphs [55]. The second layer is a Level 2 Inter-Molecular Set Transformer interaction layer, using a Self-Attention Block (SAB) and Pooling by Multihead Attention (PMA) to model non-linear blending dynamics [28, 57]. This interaction layer is evaluated directly against two baselines that combine the same molecule embeddings without an interaction step: mean pooling, and concatenation as reproduced from an existing published pairwise model.

The dataset pipeline merges single-molecule structural and olfactory data from GoodScents and Leffingwell, extracted via the Pyrfume library in Python [7], with a benchmark dataset of catalog-recorded binary chemical blends from Sisson et al.’s study. Model optimization uses a two-phase protocol: pre-training the Level 1 GAT on individual aroma-chemicals to learn structural-olfactory baselines, followed by end-to-end joint fine-tuning on binary mixtures. Optimization uses an Asymmetric Loss function to mitigate the severe label sparsity and class imbalance known to affect multi-label olfactory data [19].

Because the blend labels are compiled from catalog recommendations rather than a controlled sensory panel, their reliability is measured directly rather than assumed. Each blend descriptor is

graded into a reliability tier – Gold, Silver, or Bronze – based on (a) reciprocal agreement between the two catalog records of the same blend pair, and (b) cross-catalog agreement between GoodScents and Leffingwell on the parent molecules’ descriptors, measured using Cohen’s kappa. All three combination methods are evaluated on the resulting high-confidence (Gold-tier) test partition, trained separately on the full dataset and on Gold-tier blends only, to test whether the model comparison holds as label confidence changes.

A leakage-resistant evaluation split is used to assess the model’s out-of-distribution (OOD) generalization [13]. The chemical space is clustered into structurally related families using UMAP dimensionality reduction on 2048-bit Morgan fingerprints (ECFP4), followed by K-Means clustering on the resulting 2D projection [16], so that no test molecule’s structural family appears in training. Performance is compared across the three combination methods using Hamming Loss, Macro and Micro F1-scores, and Area Under the Precision-Recall Curve (PR-AUC), with paired bootstrap confidence intervals. Each combination method’s sensitivity to label noise is additionally evaluated by retraining on the Gold-tier partition with controlled noise injected at 0%, 10%, 20%, and 30%, and comparing the resulting decline in macro F1 across models.

This study is limited to two-molecule blends; concentration or mixing-ratio information is not modeled, and every blend is treated as an unweighted pair. Labels are binary (present or absent) rather than intensity ratings. No new human sensory panel is recruited; label reliability is instead assessed through cross-catalog and reciprocal agreement, as described above. This study does not evaluate whether Level 2 attention weights correspond to documented receptor-level antagonism mechanisms, nor does it develop a standalone ingredient-substitution or screening application; both are noted as potential directions for future work.

1.5 Definition of Terms

Aroma Molecule / Molecular Graph

A volatile organic compound that activates olfactory receptors, represented computationally as an undirected, attributed graph: atoms as nodes and covalent bonds as edges, each described by a short list of chemical properties.

Scent Descriptor (Label)

A word from a fixed vocabulary (e.g., woody, floral, citrus) naming a scent quality. Each molecule

or blend has a present/absent value for every descriptor, drawn from a standardized vocabulary of up to $L = 74$ notes.

Two-Molecule Blend

A combination of two aroma molecules treated as an unordered pair, without a chemical reaction between them: blend $\{A, B\}$ is the same as blend $\{B, A\}$.

Ground Truth (Reference-Catalog Label)

The descriptors recorded for each blend in the dataset used for training and testing, sourced from the GoodScents catalog’s recommended blending partners rather than a controlled sensory panel. This study treats these labels as a proxy for perceived scent and measures their reliability before relying on them.

Reciprocal Agreement

A measure of within-catalog label reliability for a blend pair: whether both molecules independently list each other under the same scent descriptor on their own catalog pages, versus only one molecule making the claim.

Cross-Catalog Agreement (Cohen’s Kappa)

A statistic measuring how consistently two independently maintained catalogs (GoodScents and Leffingwell) assign the same descriptor to the same molecule, correcting for the agreement expected by chance alone.

Reliability Tier (Gold, Silver, Bronze)

A per-descriptor classification of label trustworthiness. A descriptor is Gold if it shows both reciprocal agreement and adequate cross-catalog kappa, Silver if it meets only one condition, and Bronze if it meets neither.

Graph Attention Network (GAT)

A neural network that reads a molecular graph by passing information between connected atoms, learning attention weights that indicate how much each neighboring atom or bond contributes to the molecule’s overall representation.

Molecule Embedding

The fixed-length vector (here, 256-dimensional) that the Level 1 encoder produces to summarize one molecule.

Averaging and Concatenation Baselines

Two linear ways of combining two molecule embeddings into one blend representation: element-wise averaging, or placing them side by side. Because both operations are order-independent by construction, they can only preserve information shared symmetrically between the two molecules, discarding relational information about how the two molecules specifically differ from or relate to one another.

Set Transformer

A network for combining unordered collections of embeddings, built from a Self-Attention Block (SAB), which lets each molecule’s embedding be adjusted according to its partner, and Pooling by Multihead Attention (PMA), which compresses the resulting set into one fixed-length blend embedding. Unlike averaging or concatenation, its output can capture interaction between the two molecules while remaining order-independent.

Leakage-Resistant Evaluation Split

A train/test partitioning protocol in which molecules are first clustered into structurally related families, using UMAP dimensionality reduction on Morgan (ECFP4) fingerprints followed by K -means clustering, so that entire families are held out together. This tests generalization to structurally unfamiliar molecules, rather than allowing near-duplicate molecules to appear on both sides of the split.

Asymmetric Loss (ASL)

A multi-label loss function that down-weights the contribution of easily classified negative labels, preventing the model from being dominated by the many absent descriptors and keeping gradient signal focused on the rare positive ones.

Evaluation Metrics

Macro F1 averages the F1 score across descriptors, giving rare descriptors equal weight; Micro F1 pools predictions globally across all descriptors; PR-AUC measures ranking quality across classification thresholds; Hamming Loss is the fraction of misclassified present/absent decisions across all

descriptors.

Chapter 2

Review of Related Literature

This chapter discusses the relevant studies and systems that support this thesis. First, it tackles seminal research on computational olfaction, followed by the science behind the perception of scents and how different scents behave when mixed. It also examines the application of machine learning to scent prediction when the molecular components in a blend are viewed as a set. Next, the chapter explains the biological architecture beneath these interactions. It reviews hierarchical Graph Neural Networks (GNNs) and their use in the computational modeling of these biological and chemical processes, and presents the literature that justifies the proposed evaluation methodology. Furthermore, it contextualizes the justification for this research, specifically its contribution to sustainable fragrance design, given the ongoing challenges in the domain. This chapter concludes by synthesizing the information from the gathered literature to identify the specific research gap this thesis addresses.

2.1 Computational Olfaction and Single-Molecule Prediction

This first section explores the evolution of computational olfaction from early QSOR modeling with manual feature engineering to more modern graph-based methods. It reviews how these frameworks were used in predicting scent from a single molecule and the limitations of such when applied to the multi-molecule blend prediction task this thesis aims to achieve.

2.1.1 History of QSOR Modeling

Quantitative Structure-Odor Relationship (QSOR) modeling refers to the prediction of the odors and olfactory properties of molecules through the characterization of their structure [15, 53, 58]. It uses the application of chemistry, neuroscience, biology, and psychology to uncover the mechanics behind odor perception, and has long been regarded as one of the key challenges in olfactory research in machine learning [7, 31, 55]. QSOR is especially important in fields such as the fragrance and flavor industries, as it could help streamline the formulation and development of new compounds computationally, as opposed to traditional experimental wet lab methods, which can be both time-consuming and expensive [20, 39].

Early approaches in QSOR involved the manual creation of numerical features that describe and quantify the properties of a molecule [3, 15, 59]. These included molecular fingerprints, which encode the presence or absence of substructures in a binary vector, and physicochemical descriptors computed with dedicated software [3, 11, 41, 52].

Suh et al., for instance, compared classical molecular descriptors, functional-group fingerprints, and Morgan fingerprints across random forest, XGBoost, and LightGBM on a curated set of 8,681 fragrance compounds, finding that the Morgan-fingerprint models outperformed the descriptor-based ones [46]. Achebouche et al. predicted both the odors of molecules and the olfactory receptors they activate using a set of 5,955 molecules, and found that graph convolutional representations outperformed general molecular descriptors and fingerprints [1]. Saini and Ramanathan demonstrated the use of both Morgan fingerprints and Mordred descriptors, paired with a random forest classifier with multi-label support, to predict odor from molecular structures [39]. Wen et al. also used approximately 5,000 Dragon descriptors with RF, SVM, gradient boosting decision trees, multilayer perceptron, XGBoost, and LightGBM, comparing them against GNN approaches to also predict odor [51].

The common denominator across all of these approaches is that they convert the molecule, specifically its properties, into fixed, handcrafted numerical representations, which are then fed into classical ML models. Recent work such as Suh et al., Achebouche et al., and Wen et al. continues to include descriptor-based classical models, but mainly as baselines to quantify how much fingerprint- and graph-based learning improves upon manual feature engineering [1, 46, 51]. While effective, these methods across both eras require the manual selection of descriptors in advance [4, 27, 59]. This means the model is not actually able to determine on its own which structural features are

relevant for the task it is trying to solve. This necessitated the advancement to graph-based deep learning methods, which improve the odor prediction pipeline by teaching models to learn task-relevant features directly from structure [2, 4, 18, 58] which will be discussed in the next section.

2.1.2 The Principal Odor Map

Addressing the aforementioned limitation is Lee et al.’s Principal Odor Map (POM), which maps a molecule’s chemical structure to its smell using Graph Neural Network (GNN) techniques [27]. Rather than evaluating a molecule’s scent using a static representation built from predefined hand-crafted descriptors, POM learns directly from the molecule’s structure during training. It does this by treating each molecule as a graph, where nodes and edges represent atoms and chemical bonds, respectively; each node and edge is itself a vector, with nodes storing properties such as atomic number, and edges storing bond properties.

Through message passing, the GNN dynamically assigns weights to each of the molecule’s structural features across multiple passes, producing a numerical representation of the molecule. The final layer predicts the molecule’s specific odor qualities, while the penultimate layer forms the actual Principal Odor Map, which the paper itself describes as a high-dimensional, data-driven representation that is able to preserve human perceptual distances and hierarchies, positioning molecules with similar smells closer together in the generated odor space [27].

The setup was as follows: Lee et al. used approximately 5,000 molecules derived from merging data from the GoodScents and Leffingwell fragrance databases to produce a multi-label output of 138 distinct semantic odor descriptors. It was then validated against a human benchmark, wherein approximately 15 trained panelists rated molecules using a narrowed-down set of 55 odor descriptors. The results show it was able to match the trained panel’s verdicts on every label except for musky scents, and was even able to surpass the median panelist on 30 out of the 55 validation labels (55%) [27].

This demonstrates the relative effectiveness of GNNs in predicting the scent of a molecule from structure alone. However, despite this significant advancement, these results were derived exclusively from single, unmixed molecules.

2.1.3 Limitations of Single-Molecule Models

Despite GNNs like POM being able to achieve human-like performance in scent prediction, the fact that they only operate on the scale of a single molecule demonstrates a fundamentally limited architecture that fails to reflect the reality that odorants rarely appear in isolation [9, 27, 49, 56]. Real-world scents are rarely composed of single molecules acting on single receptors; rather, they are blends of molecules that can contain individual components numbering up to the hundreds [17, 56]. Consequently, when a method focuses only on a single isolated graph, it is unable to predict new scents that may emerge from the mixture of two or more compounds.

Lee et al. admit this limitation themselves, noting that while POM uses a coordinate system to categorize scents, it alone cannot determine the behavior and outcomes of different molecules when mixed [27]. Moreover, single-molecule coordinate systems do not reflect a biologically plausible architecture, failing to capture the biological complexity of how the mammalian peripheral olfactory system actually operates [8, 21].

2.2 Psychophysics of Odor Mixtures

The unsuitability of single-molecule models for odor prediction in multi-molecular blends signifies the need for a similarly multi-molecular model. This section discusses the groundwork for predicting the behavior of different molecules when mixed, specifically the application of psychophysics to machine learning: how blends deviate from their original descriptors, how the brain processes them, and the concept of linearity in odor perception that motivates testing a Set Transformer against naive pooling.

2.2.1 Scent Blending Mechanics

A paper released by Sisson et al. introduces a labeled dataset consisting of molecule pairs to measure the divergence of the emerging scent when two molecules are mixed from their original scent labels, using two metrics: union and intersection. The union score operates under the assumption that the emerging scent will smell like any of the scents either parent molecule has (additive mixing). On the other hand, the intersection score operates under the assumption that the outcome of the two molecules when mixed will only come from the scents the parent molecules share (subtractive mixing) [43].

The Jaccard results show 0.12 for union and 0.24 for intersection [43]. These low scores indicate that neither additive nor subtractive mixing reflects the real behavior exhibited by molecule pairs when mixed, according to catalog blend labels. This shows that the naive use of linear math cannot properly replicate the unpredictable behavior that molecule pairs exhibit when mixed.

This non-linearity can be characterized by the phenomena of masking/suppression and emergence, wherein the former describes what happens when a scent present in either parent molecule fails to appear in the blend, while the latter describes what happens when a new scent is perceived that was not present in either parent molecule’s odor descriptors. The results of Sisson et al. show that, on average, 1.2 notes are suppressed and 0.2 notes emerge per blend. Furthermore, an odds-ratio analysis reveals that certain scents, such as "eggy" or "cabbage", were more likely to emerge, while certain scents like "peach" or "fresh" were more likely to be suppressed [43]. This demonstrates that these two phenomena follow a certain learnable pattern, and that the behavior is not merely random, which motivates using a model that can learn these interactions, such as a Set Transformer.

2.2.2 Elemental vs. Configural Processing

Following the examination of the non-linear behavior of the scents of molecule pairs when mixed, this section will provide a sort of basis through the lens of how humans process odors. Studies by Romagny et al. and Coureaud et al. describe two strategies that the human brain uses to process smells, namely configural odour perception and elemental odour perception. With elemental processing, the brain distinguishes the different components that make up a scent. On the other hand, with configural processing, the brain perceives a new scent that is distinct from the original scents that make up its components [10, 38].

To better illustrate their difference, one could imagine two molecules, molecule A and molecule B, which in isolation smell like scent A and scent B respectively. Elemental odour perception would allow a human to smell at least one of, or both, scent A and scent B, while configural odour perception would allow a human to smell a scent that is neither scent A nor scent B.

Which of the two is perceived can depend on the proportions of the components, which is one reason this study, which does not model concentration, treats each blend as an unweighted pair.

All of this suggests that a multi-molecule odor prediction model should be able to learn how molecules interact, since emergent scents cannot always be described by the individual components alone. This motivates the Set Transformer used in this study.

2.2.3 Perceptual Linearity vs. Receptor-Level Non-Linearity

Not all evidence points to strongly non-additive mixing. Pellegrino et al. [34] had a trained panel rate 432 mixtures using a 51-term list and found that a linear average of the components’ ratings predicted mixture profiles about as well as the noise in the panel’s own ratings allowed. They argue that apparent emergence often reflects the limits of describing an odor with a few words. Binary catalog labels, as in Sisson et al. [43], instead record note loss and note appearance in many blends. Whether an interaction step helps for binary multi-label prediction is therefore an open question, which this study tests.

In any case, in simpler terms, odors could be considered linear when viewed through a continuous scale, while non-linear when viewed through a binary lens. Our thesis specifically takes the angle of the latter to be able to produce multi-label predictions for odors, similar to Sisson et al.’s [2025] work, which is why a Set Transformer is tested for this prediction task.

One proposed reason a note may disappear in a blend is competition at olfactory receptors: an odorant that binds a receptor without activating it can block another odorant from doing so [14, 22]. This study does not model receptors; the point is only that suppression has a physical basis.

2.3 Machine Learning on Sets and Chemical Mixtures

Having established the baseline information on computational olfaction and the behavior and psychophysics of odorant molecules when blended together, this next section discusses the machine learning perspective on the problem, specifically the mathematical foundation underlying the treatment of resulting scent compounds as sets and the relevant prior work on the topic.

2.3.1 Permutation Invariance on Sets

A combination of two or more molecules could be viewed as a mathematical set. An accurate depiction of a fragrance blend of these molecules requires that they be presented in such a way that their positioning in that list is arbitrary, not sequential [43, 49, 58]. For instance, assume the combination of molecule A and molecule B produces scent C. It should be of no relevance whether it is molecule A + B or molecule B + A; it should always produce scent C. For that reason, it is important for any mathematical model used to predict the resultant smell from the combination of different molecules to follow this principle, which is called permutation invariance.

Zaheer et al. [57] established the math of permutation invariance and introduced Deep Sets, a deep learning framework that uses a neural network designed to process datasets in a non-sequential manner for unordered data, which serves a different function from architectures that must process data in order (e.g., molecular compounds vs. protein sequences). This is referenced as the first of two properties they introduced in their study describing permutation invariance, positing that for a function acting upon a set, it must be indifferent to the order in which each element appears, mathematically formulated as $f : 2^X \rightarrow Y$ for any permutation, i.e., for any permutation π :

$$f(\{x_1, \dots, x_M\}) = f(\{x_{\pi(1)}, \dots, x_{\pi(M)}\}).$$

Their second theorem explains how to make the first property possible, stating that the function must be able to individually transform each element ϕ , the results of which are aggregated and then transformed again to ρ , mathematically formulated as decomposable to:

$$\rho \left(\sum_{x \in X} \phi(x) \right),$$

which guarantees the permutation invariance of a set [57].

However, as can be seen in the second theorem, the transformation of each element is done before pooling and is processed in isolation. Therefore, the information about how individual elements may interact with each other is not captured and gets lost, which is pointed out by Lee et al [28]. This is a critical issue in the context of fragrance blends specifically because of the non-linear behavior that odor molecules exhibit. The non-linear behaviors like emergence and masking are not captured if the molecules are treated as having no bearing on each other, which may not hold for real blends.

To bridge this gap, Lee et al. propose the use of a Set Transformer, where instead of processing the elements of a set individually, they apply self-attention mechanisms they call SAB (self-attention block) in the processing of the elements, wherein the different elements are able to learn from each other, such that the extracted features to be used for pooling already contain useful information about the elements' relationship with each other. Moreover, they also developed PMA, or Pooling by Multihead Attention, which replaces the second step in the Deep Sets architecture wherein, instead of fixed sum or mean pooling, PMA also employs self-attention mechanisms in the pooling itself, such that it is able to pick out which of the elements bear more weight for the task at hand. Both of these innovations directly apply to our thesis problem. With the use of SAB, information about the interaction between odorant molecules is made available to the model, and made meaningful with the use of PMA.

Blend models may also concatenate the two embeddings side by side, as in the pairwise model of Sisson et al. [43]. Concatenation keeps both embeddings in full, but its output depends on which molecule is listed first.

2.3.2 Precedents in Mixture Modeling

Zaheer et al. [57] provided the groundwork for permutation invariance in sets, while Lee et al. [28] extended it with self-attention, a general set method that later studies applied to molecular mixtures. Plenty of studies have made use of their work in different realms of chemistry.

For example, Qin et al. [35] built the SolvGNN architecture, which aimed to predict the activity coefficient between binary and ternary chemical mixtures by capturing their intermolecular interactions at two levels. First is the atomic-level (local) graph convolutions, which happen in the individual molecules, where features are extracted from each component atom, including the edges that connect them. This structural information is propagated and embedded in preparation for the next level, which is the molecular-level (global) message-passing graph, which operates at a broader scale as the nodes are now the molecules themselves, with the edges representing molecule-to-molecule interactions, such as hydrogen bonding. In this complete graph, message-passing algorithms are employed, using a Gated Recurrent Unit (GRU) to update each node with each pass, to capture mixture compositions and non-ideal fluid behaviors, with Qin et al. [35] arguing that these dynamic measures are essential to explicitly model the real physical intermolecular forces in mixtures that naive pooling fails to do.

Similarly, Sanchez Medina et al. [41] introduce the Gibbs–Helmholtz Graph Neural Network (GH-GNN), which also predicts activity coefficients of chemical mixtures, but at different temperatures. What differentiates GH-GNN’s architecture from SolvGNN’s is that, instead of having two separate levels for GNN processing, GH-GNN has a single shared GNN, wherein the nodes and edges already contain both local and global vector features in a pair, which get updated over multiple layers of message passing. Afterwards, a global mean pooling operation is done to compress the data from each graph to create a vector embedding, concatenated with the updated global feature to create the final molecular fingerprint, which will then be transferred to another graph called the mixture graph, which maps out the hydrogen-bonding interaction between the solvent and solute molecules. Sanchez Medina et al. [41] defend this design by explicitly building the temperature-dependency relationship into the model’s structure through a Gibbs–Helmholtz-derived thermodynamic expres-

sion, rather than relying on a generic GNN to infer it implicitly from data alone. This closely mirrors, and inspired, our thesis’s approach of doing the same shared single-molecule encoder and then a separate mixture-level interaction step, but with odorant pairs instead of solvent and solute.

Lastly, Tom et al. [49] proposed POMMix, a more relevant precedent for our topic, as it is in the field of olfactory perception, aiming to predict the similarity of the scent of two mixtures. Theirs employs a 3-layer architecture, with the first making use of the previously discussed POM, where a GNN processes the atomic bond traits of the molecules along with the scent labels to create a multi-dimensional vector that describes the scent of the molecule. The second layer employs a module they call CHEMIX, which uses a self-attention mechanism across the graph from the first layer, doing set-pooling to produce a permutation-invariant mixture embedding that is able to tell which components contribute to the nonlinear behaviors that odorants exhibit. The last layer of their architecture is the cosine predictive head, which evaluates the similarity of mixtures according to their scent by calculating the scaled cosine distance between their respective embeddings. Their ablation study compared several molecular embedding types including handcrafted RDKit descriptors and pretrained language-model embeddings, combined with both a classical model (XGBoost) and their attention-based CHEMIX module, finding that embeddings derived from POM consistently outperformed the alternatives, reinforcing that structure-aware, attention-based representations capture mixture behavior more effectively than simpler alternatives.

All three studies tackle three different prediction problems, yet use the same underlying concept of dynamic self-attention mechanisms, as they prove to perform better and better serve the problem of being able to model molecular interaction, as compared to basic aggregation and naive pooling/concatenation, going on to show the importance of using this framework for this paper, which aims to capture intermolecular interactions between odor pairs.

Hierarchical GNNs have also been used for property prediction within a single molecule, moving from atoms to fragments, as in HiGNN and HimGNN [18, 59]. POMMix, SolvGNN and GH-GNN, in contrast, operate across molecules. The present study follows the second kind: a shared single-molecule encoder followed by an interaction step between the two molecules.

2.4 Olfactory Datasets and Label Reliability

This section will discuss the issues related to olfactory data, beginning with the unsuitability of community-driven scent databases for modeling purposes, then discuss the limitations of single-molecule reference catalogs, and finally, the problem of label overlap and the lack of a standard descriptor vocabulary.

2.4.1 Reference Catalogs

The quality of sources for public scent data are not equal. Community platforms such as *Fragrantica* and *Basenotes* describe finished commercial perfumes in the words of their users, which makes them unsuitable for learning how individual chemicals relate to scent[7, 52]. Because of these, researchers use catalogs such as *GoodScents* and *Leffingwell* for reference, as they record descriptors for individual aroma chemicals and have been used to train state-of-the-art prediction models[27, 43]. The *Leffingwell PMP 2001* database was compiled by olfactory experts, usually practicing perfumers, and its free-text descriptions were standardized into 113 descriptors by Sanchez-Lengeling et al.

2.4.2 The Blended-Pair Dataset

Sisson et al. [43] compiled blend labels from the lists of suggested blending partners on *GoodScents*, which group each partner under an odor heading. Their crawl was able to produce more than 160,000 pairs and, after standardization and filtering, resulted in up to 74 odor labels. Each molecule’s page was crawled separately, so a pair can be listed from both molecules’ pages; their pair-building code counts these listings but does not keep the count. The dataset has no concentration information, and its labels are catalog suggestions rather than measurements from a sensory panel.

2.4.3 Human-Rated Datasets

Expert-compiled catalogs such as *Leffingwell* are assigned by people, but they are used here as training data and cannot serve as an independent check. Independent human-rated data are therefore needed. Human-rated scent datasets exist for single molecules and for mixtures, but to our knowledge none that gives scent descriptors for two-molecule blends is publicly available at the time of writing. Bierling et al. [6] provide yes-or-no ratings of 16 scent words for 74 single odors from about 1,200

laypeople. Pellegrino et al. [34] collected trained-panel ratings of mixtures with a 51-term list and have stated that their data will be released after publication [personal communication].

2.4.4 Absence of a Standard Vocabulary

No standard set of odor descriptors exists. Catalogs and studies use vocabularies of different sizes, and descriptors overlap in meaning[43]. To organize descriptors, Sajan et al. [40] published an expert-curated taxonomy that groups 557 source-based descriptors from the Pyrfume repository into 16 scent families; this study reports results per family in addition to per label (Section 3.1.6).

2.5 Out-of-Distribution Splitting Strategies

This section will be the last technical segment of this chapter, providing a review of the standard data-splitting strategy of scaffold splitting, and an examination of how its application falls short in sparse chemical spaces, which then motivates the need for more rigorous out-of-distribution (OOD) evaluation. This section then grounds and discusses the clustering-based alternative our proposed framework will employ.

A common method to partition data for training, testing, and validation in chemoinformatics and molecular machine learning is scaffold splitting [16, 45]. In this method, molecules are represented by graphs of their structure, and the core framework of these graphs is referred to as their “scaffold” (Bemis–Murcko scaffolds) [54]. Molecules that share the same scaffold are grouped together, and these groups are then distributed among the training, testing, and validation sets to ensure that the test set contains structurally different molecules from the training set, which is a basic form of out-of-distribution (OOD) evaluation, which as opposed to random splitting, simply distributes molecules randomly across training, validation, and test sets [13, 16, 43, 45].

The caveat with this method is that, when the dataset is sparse, it collapses toward random splitting. For instance, similarly structured molecules get fragmented across both the train and test splits, leading to severe data leakage and inflated test scores as scaffolds in standard benchmarks could be populated by only one or two molecules each, undermining the grouping effect the method depends on [13]. Another problem demonstrated by scaffold splitting is that even molecules that are near-identical (as computed by Tanimoto similarity) [16, 45], even differing by as little as a single atom, can, under this method, end up in opposite splits, as observed by Guo et al. and Steshin et al.

[16, 45], who argue that this problem causes models to “memorize” rather than generalize from the data. Because of these problems, scaffold splitting can misleadingly produce scores that are overly optimistic.

Shimakawa et al. [42] propose doing topological pre-clustering by projecting 2048-bit ECFP4 fingerprints onto a 2D map using 2D Uniform Manifold Approximation and Projection (UMAP), then grouping them together using spectral k -means clustering. This was then validated using Leave-One-Cluster-Out Cross-Validation (LOCOCV), where clusters are left out in rotation for training and testing, such that what the model was trained on does not show up in what it is tested on, meaning the model actually learns, rather than simply memorizing the data it was trained on, which is important for OOD evaluation.

However, while these methods may work well for single-molecule data splitting, they would be insufficient for multi-molecular fragrance mixtures, because an individual molecule can be a component of varying other molecules in different blends, meaning a molecule seen during training could reappear as a component in a different, unseen test blend, leaking information about that molecule across the split, causing the performance metrics to inflate. Therefore, Sisson et al. [43] propose “graph carving”, using a meta-graph to describe the pairwise/multi-molecule blend dataset, where each node represents an individual molecular graph and each edge represents a tested blend combination. Edges connecting a molecule from one partition to a molecule in another are discarded to guarantee no component overlap between splits. In their original implementation, this partitioning was done by randomly assigning molecules to each side and repeating until every label had at least one training and one test example, since a true scaffold-based split was not compatible with their sparse label distribution.

Building on this, our proposed framework instead applies the graph-carving protocol on top of the UMAP + k -means clustering discussed above, rather than random or scaffold-based partitioning. By applying this combined methodology to our merged GoodScents–Leffingwell–Sisson dataset, clustering molecules via UMAP and k -means, then carving the resulting blend meta-graph along those cluster boundaries, our proposed framework is evaluated on a strictly out-of-distribution test set.

2.6 Synthesis and Research Gap

2.6.1 Synthesis of Literature

Current models in computational olfaction can successfully map single-molecule structures to their appropriate sensory descriptors [27]. However, because these compounds are evaluated in isolation, nonlinear psychophysical interactions such as masking and emergence—which determine what the final resulting scent would smell like when multiple odorant molecules are mixed—are not taken into account [22, 38, 43]. Human-panel evidence on mixtures is mixed: continuous ratings are well predicted by a linear average of the components[34], while binary catalog labels record note loss and note appearance in many blends[43].

From an architectural standpoint, permutation-invariant set processing must be used to predict the behavior of multi-molecule mixtures [57, 28]. Simple linear pooling (mean or sum) can make different pairs look identical, because pairs with the same average produce the same combined representation [47].

Furthermore, standard scaffold splits suffer from severe data leakage at the component level in mixture spaces [13]. As a result, continuous manifold clustering must be used in conjunction with meta-graph carving to guarantee leak-free evaluation [42, 43]. Finally, the blend labels in public data come from catalogs rather than sensory panels, and their reliability is rarely measured.

2.6.2 Statement of the Research Gap

No published study has compared an attention-based set model, the Set Transformer, against mean pooling and concatenation for predicting the scent labels of two-molecule blends, evaluated on a reliability-graded subset of the labels and tested for sensitivity to label noise.

2.6.3 Positioning of the Present Study

To close this gap, this study presents a Structure-Aware Hierarchical Graph Neural Network (SA-HGNN) that applies a weight-shared Level 1 Graph Attention Network (GAT) to encode information on a per-molecule basis, and makes use of an explicit Level 2 Set Transformer (SAB + PMA) to handle the interactions between molecules, thereby allowing the model to represent interactions between the two molecules. This framework is evaluated on a leakage-resistant Morgan-UMAP

k-means graph-carved benchmark to quantify the value of explicit set attention over naive pooling; the label-consistency audit and the noise test show how far the conclusions depend on the labels. By reducing dependence on physical trial-and-error formulation, models of this kind may support SDG 12, and the open reliability audit supports SDG 9

Chapter 3

Theoretical and Conceptual Framework

3.1 Theoretical Framework

This section presents the theory behind the SA-HGNN architecture and the label reliability analysis. It first describes how each aroma molecule is represented as a graph of atoms and bonds. It then explains Graph Attention Networks (GAT), which encode each molecule, followed by set invariance and Set Transformers, which combine the two molecules of a blend. The fourth part explains why simple averaging can lose information about how the two molecules relate to each other, which motivates testing an interaction layer against averaging and concatenation baselines. Because this argument describes a worst case and not an observed result, whether the interaction layer improves prediction is treated as an empirical question. The final part covers the measures used to assess label reliability: reciprocal agreement, Cohen’s kappa, and learning from noisy labels.

3.1.1 Graph-Theoretic Molecular Representation

Each aroma molecules is modeled as an undirected, attributed molecular graph in which atoms are nodes and bonds are edges, and create a mathematical representation where atomic and bond features are updated via message passing between the connected atoms [55]. This would allow the Graph Attention Network (GAT) to extract structural scent profiles regardless of atom ordering.

In cheminformatics, chemical compounds are often written as one-dimensional character strings, such as text-based Simplified Molecular Input Line Entry System (SMILES) strings. However, these text-based strings are syntax-sensitive. A single molecule can be written as many as single valid string depending on the starting atoms, and small edits can produce a completely different molecule [33, 52]. Representing a molecule as a graph captures its bond connectivity without depending on the order in which atoms are listed, which matches how the Graph Attention Network passes information between bonded atoms.

Mathematical Formalism

Each aroma molecule will be defined as an undirected, attributed graph $G = (V, E)$, where:

- V represents the finite set of $N = |V|$ nodes, corresponding to the non-hydrogen atoms in the molecule. Hydrogen will be treated as implicit properties embedded directly within the feature vectors of their parent heavy atoms, rather than independent nodes, which will reduce the graph complexity and computational overhead [4].
- E represents the set of $M = |E|$ undirected edges. An edge $(v_i, v_j) \in E$ represents a covalent bond between heavy atoms v_i and v_j . Because a bond is a mutual sharing of electrons between two atoms, it has no direction: $(v_i, v_j) \in E$ if and only if $(v_j, v_i) \in E$ [4].

The connectivity of the molecule is described by a symmetric binary adjacency matrix $A \in \{0, 1\}^{N \times N}$, where $A_{ij} = 1$ if a bond exists between atoms v_i and v_j and $A_{ij} = 0$ otherwise [59]. In implementation, the graph is stored in a sparse format instead of a dense matrix (Section 3.3.2). Each node and edge also carries a feature vector:

- Node vector $\mathbf{x}_i \in \mathbb{R}^{46}$, which is the initial state $\mathbf{h}_i^{(0)}$ used by the GAT. It records the atom’s element, formal charge, number of bonds, valence, hybridization, aromaticity, attached hydrogens, ring membership and ring size, and chirality, in 9, 1, 6, 7, 6, 1, 5, 6+1, and 4 dimensions respectively, for a total of 46 (Table A.1) [44, 59].
- Edge vector $\mathbf{e}_{ij} \in \mathbb{R}^{10}$, which records the bond’s type, conjugation, ring membership, and stereochemical configuration (Table A.2) [59].

Olfactory Domain Mapping

Some structural details are known to affect scent, including chirality (the two enantiomers of carvone, for example, smell of spearmint and caraway), ring structure, and aromaticity. The node and edge features encode these details so that the model has access to them. Features are computed with the RDKit cheminformatics library [7, 44, 27], using feature types common in molecular and odor-prediction models [27, 44]. The features describe individual atoms and bonds. Larger structures such as functional groups are not encoded directly; the GAT can learn to recognize them by combining information across neighboring atoms. The exact composition of the 46-dimensional node feature vector and the 10-dimensional edge feature vector is given in Appendix A.

Analytical and Methodological Justification

Aroma molecules modeled as graph structures instead of physical descriptors or using the text-based SMILES strings provide key advantages:

- **Order Independence:** The encoder sees the molecule as one connected structure as a sequential list but as a whole, so the output does not depend on the sequence in which atoms are listed. This concerns atom order within one molecule and is separate from the order independence of the two-molecule blend discussed in Section 3.1.3. This ensures that it predicts scents no matter how the atoms are arranged [33].
- **Localized Feature Preservation:** Specific clusters of atoms, called “osmophores,” are thought to be responsible for particular odor qualities [44]. Since each atom carries its own features and the GAT aggregates information from bonded neighbors, the encoder can learn patterns in local chemical environments before combining them into a molecule-level embedding [55, 27]. The model is not given receptor data, so it does not identify receptor-binding sites.
- **Isomers:** Molecules with similar atoms but different bonding—*isomers*—have different graphs and are therefore distinguished. Molecules that have the exact same atoms and bondings called *stereoisomers* are distinguished if explicit attributes—chirality and double-bond configuration—are added into its features, and 2D graph does not capture 3D shape or conformation [4, 52, 58].

3.1.2 Deep Learning Theory: Graph Attention Networks (GAT)

The model uses a Graph Attention Network (GAT) to map each molecular graph to a numerical vector. Atoms are the nodes and the bonds connecting them are the edges. Instead of combining a neighboring atom’s information with fixed weights set by the graph structure, a GAT learns a weight for each neighbor, so the encoder can give more influence to some neighboring atoms than others when building its vector representation [27, 55]. This is loosely analogous to olfactory receptors responding more strongly to some structural features than to others, but the learned weights are model parameters and are not evidence of receptor behavior or chemical importance.

Let $\mathbf{h}_i^{(0)} = \mathbf{x}_i$ denote the initial 46-dimensional feature vector of atom v_i . A learnable weight matrix $\mathbf{W}$ projects the features of each atom into a hidden space of size $F' = 256$. To score how much atom v_i should attend to a neighbor v_j , a single-layer feedforward network with parameters $\mathbf{a}$ evaluates the concatenated projected features of the two atoms and the projected 10-dimensional bond feature vector $\mathbf{e}_{ij}$ between them [55, 59]:

$$\tilde{e}_{ij} = \text{LeakyReLU}\left(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \parallel \mathbf{W}\mathbf{h}_j \parallel \mathbf{W}_e\mathbf{e}_{ij}]\right), \quad (3.1)$$

where $\parallel$ denotes concatenation. Including $\mathbf{e}_{ij}$ lets the score depend on the bond type, conjugation, ring membership, and stereochemical configuration. The LeakyReLU, with a negative slope of 0.2, keeps a small gradient for negative scores so that they do not become stuck at zero [55].

Atoms have different numbers of neighbors, so the raw scores are normalized across the neighborhood $\mathcal{N}(i)$ of atom v_i with a softmax, so that the weights for each atom sum to one [55, 4]:

$$\alpha_{ij} = \text{Softmax}_j(\tilde{e}_{ij}) = \frac{\exp(\tilde{e}_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\tilde{e}_{ik})}. \quad (3.2)$$

The normalized weights α_{ij} scale the projected neighbor features, and the Exponential Linear Unit (ELU) activation is applied to give the updated representation of atom v_i [55]:

$$\mathbf{h}_i^{(l+1)} = \text{ELU}\left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{h}_j^{(l)}\right). \quad (3.3)$$

To stabilize learning, the encoder runs $K = 4$ independent attention heads in parallel. In the hidden layers their outputs are concatenated, and in the final layer they are averaged [55]. Global mean pooling over the atoms of the molecule then gives the final 256-dimensional molecular embedding.

3.1.3 Set Invariance and Set Transformers

Mixing two fragrance ingredients regardless of their order is presumed to smell the same. Mathematically, mixing Molecule A and Molecule B will have the exact same scent as mixing Molecule B and Molecule A. The blend follows the commutative property, which means that it behaves as an unordered set where the order of input does not change the output. A blend is therefore treated as an unordered set of two molecules rather than an ordered pair, and the function that combines them is required to be permutation-invariant, so that its output does not depend on which molecule is listed first [43, 57].

The Deep Sets Formulation

Deep Sets addresses this order-independence problem. It shows that this framework guarantees permutation invariance by processing each set element through an identical neural network transformation before aggregating their representations using a symmetric function, such as summation and averaging [57]. Formally, a function $f(X)$ operating on a set of molecular embeddings X is defined as:

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right). \quad (3.4)$$

The function ϕ represents an encoder network that projects each individual molecular embedding x to latent representation, while the sum operator aggregates these individual representations regardless of the order they are summed in [57]. In this study, ϕ corresponds to the Level 1 GAT encoder, which produces a 256-dimensional embedding for each molecule independently; summation or averaging over the two resulting embeddings is one candidate for the aggregation step ρ applies to, evaluated here as the averaging baseline. A limitation of this formulation is that ϕ evaluates each molecule in isolation, so Deep Sets, by construction, assumes the two molecules do not affect each other. Real-world olfactory mixtures can exhibit psychophysical phenomena such as masking and emergence that this formulation has no mechanism to represent [43, 47].

Set Transformers and Attention-Based Pooling

Simply averaging or concatenating the embeddings of the two molecules, as described above, combines them through a fix operation that does not depend on either molecule’s structure. To let the two embeddings influence one another, a Set Transformer allows each representations to interact

while remaining permutation-invariant [28]. The Set Transformer is composed of two blocks. The set Set Attention Block (SAB) lets the two molecule embeddings in a blend to cross attend to one another [28, 49]. This let each representation to condition on its pair before pooling. This block is intended to give the model a mechanism for representing psychophysical phenomena such as masking and emergence, which averaging and concatenation fail to express [35, 43]; wheter it does so in practice is tested empirically in this study.

The second block is Pooling by Multihead Attention (PMA):

$$\text{PMA}_k(X) = \text{MAB}(S, \text{rFF}(X)). \quad (3.5)$$

This process begins by taking the molecular embeddings $\mathbf{X}$ and passing them through a row-wise feedforward network that prepares each molecule’s data for the final pooling calculations. The PMA block introduces a set of k learnable seed vectors, denoted as (S) . These vectors are refined during training to actively extract the dominant features of the scent blend. The Multihead Attention Block (MAB) lets these seed vectors (S) attend to the polished molecule embeddings $(\text{rFF}(X))$, compressing the set of two molecule embeddings into a fixed-size blend representation [28, 43].

3.1.4 Readout Expressivity Limits and Symmetry Collapse

The two chemical embeddings must be merged into a single representation before classification happens. Standard Graph Neural Network (GNN) designs are usually simple when merging vectors in which it typically use linear permutation-invariant pooling, such as sum or mean pooling [57]. This kind of pooling has a mathematical ceiling: many different pairs of embeddings can share the same average, so a linear readout cannot always distinguish them, regardless of how much structural detail the encoder captured beforehand [47]. This is a limit on what a linear readout can represent in the worst case, not a claim about how this specific model performs on this dataset [43]. This argument concerns the mixture-level readout, where two molecule embeddings are combined; the within-molecule mean-pooling readout in Level 1 follows common practice in molecular GNNs and is not the focus of this study.

Why Averaging Loses Information

Averaging two molecule embeddings can make structurally different pairs indistinguishable to the model. In one dimension, the pairs (2, 8), (4, 6), and (5, 5) all average to 5, so after averaging, the

model cannot tell them apart no matter how precisely each molecule was encoded. Applied to blend prediction, two chemically different molecule pairs can average to the same blend representation, even when the GAT encoder produced embeddings that were genuinely sensitive to the difference between them. Concatenation avoids this specific collapse but is not permutation-invariant on its own, since its output depends on which molecule is listed first. This is the gap the Set Transformer is intended to close: letting the two embeddings interact before combining them, while remaining order-independent, unlike concatenation. A full representation-theoretic derivation of this argument, including its automorphism-group formulation, is given in Appendix A.1.

3.2 Conceptual Framework

The conceptual framework of this study operationalizes the theoretical formulations into a systematic computational pipeline structured around the Input-Process-Output (IPO) model. Figure 3.1 outlines the structural interaction between input data representations, the two-level hierarchical deep learning architecture, and the downstream analytical and practical deliverables.

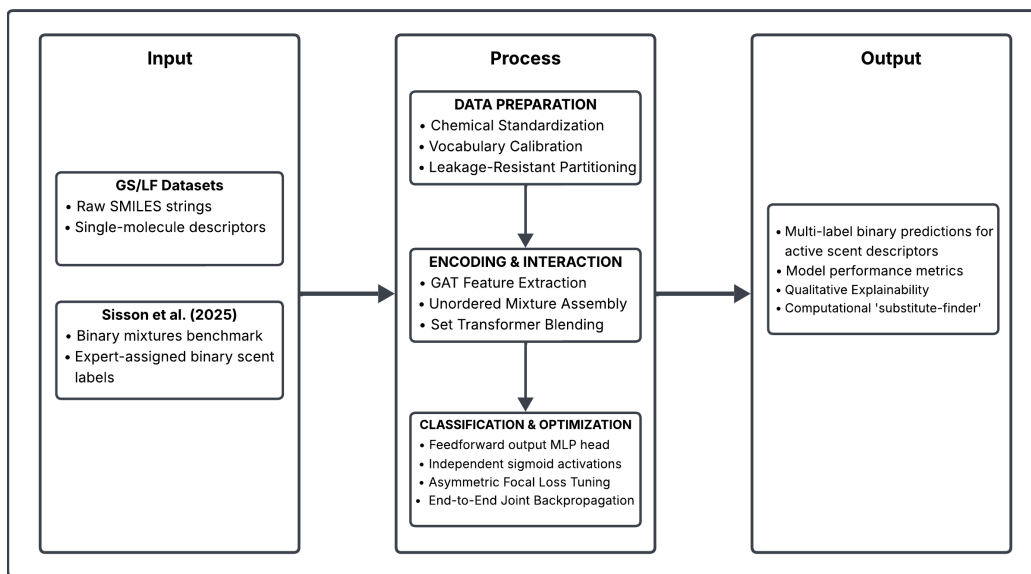


Figure 3.1: Input-Process-Output (IPO) Model.

3.2.1 Input Phase

The computational data pipeline starts with the acquisition, curation, and gathering of data from olfactory databases. The raw SMILES strings are extracted from the GoodScents and Leffingwell repositories accessed via the Pyfrume Library in Python [7]. These raw chemical blueprints are then paired with expertly annotated odor tags to establish baseline inputs required from the pre-training phases. The binary mixture profiles are retrieved from the Sisson et al. (2025) benchmark dataset [43]. The retrieved benchmark dataset provides standardized chemical pair identifiers (CIDs) and multi-label scent descriptors that will help the network to learn and predict the combined scent of two scent blends.

3.2.2 Process Phase 1: Data Preparation and Graph Carving

To raw chemical input will first pass through a standardization pipeline. Salts and other non-fragrance components are removed using RDKit’s SaltRemover module, leaving the main molecule. Compounds containing elements other than C, H, O, N, S, F, Cl, Br, and I are dropped, since these fall outside the range typical of aroma chemicals [33]. The molecular weight should be between 20 and 600 g/mol. Below this range is considered irrelevant, while above the range is considered too heavy to be perceived as a scent at all [27]. The surviving SMILES strings are canonicalized using RDKit. The scent descriptors text from the source databases is standardized by merging redundant terms and eliminating descriptors that appear rarely to learn from [9]. The blend labels are then audited for reliability before any model is trained. Reciprocal agreement is measured for each blend pair, and cross-catalog agreement between GoodScents and Leffingwell is measured on the parent molecules using Cohen’s kappa. Each blend descriptor is assigned a Gold, Silver, or Bronze tier based on these two measurements, with the tier rules fixed in advance. To test how well the network handles unfamiliar compounds, molecules are grouped into structural families by UMAP and K-Means on 2048-bit Morgan fingerprints [16]. The Graph-Carving algorithm then divides the mixture meta-graph along family clusters. This guarantees that validation and testing pairs do not contain any molecule encountered during training [13].

3.2.3 Process Phase 2: Encoding and Interaction

The second phase coonverts the flat, two-dimensional 2D molecular graphs into continuous high-dimensional vector spaces and models non-linear interactions between paired compounds. In the encoder, individual atoms and bonds are processed as graph nodes and edges [55]. Since the Level 1 GAT shares its weights, it encodes both molecules under the same chemical rules, which then guarantees symmetry [41]. After encoding and updating atomic neighborhoods, global mean pooling condenses these vector spaces into a single 256-dimensional embedding for every compound [41, 55].

The two embeddings are then combined into a single representation using one of three methods, which are compared against each other mean pooling; concatenation, reproduced from an existing published pairwise model [43]; or the Level 2 Set Transformer, in which each embedding is adjusted according to its molecular before the two are pooled into a single 256-dimensional vector [28]. The first two combine the embeddings through a fixed operation, while the Set Transformer lets the two molecules interact. Whether that interaction improves prediction is the central question of the study.

3.2.4 Process Phase 3: Classification and Optimization

Once the Level 2 yields the 256-dimensional blend vector spaces, it will then be translated into multi-label scent predictions. This Multi-Layer Perceptron (MLP) head contracts sequentially from 256 to 128 and ends with L individual outputs to match the target scent classes [43]. Because scent profiles are inherently multi-label, the classification layer utilizes independent sigmoid activation functions [39]. These activations compute individual probability scores for the joint presence or absence of each unique target scent descriptor. The system is trained end-to-end for network optimization. This means feedback from prediction errors flows backward through all layers simultaneously, fine-tuning both the Level 1 molecular encoders and the Level 2 mixture blocks guided with Asymmetric Focal Loss [19]. Each combination method is trained under two data conditions, on all reliability tiers and on Gold-tier blends only, and, for the noise sensitivity test, on Gold-tier blends with a controlled fraction of labels corrupted. Olfactory datasets are highly sparse, meaning a single scent profile contains only a few active descriptors alongside dozens of inactive ones (zeros). Under standard training rules, the model would quickly learn that it can achieve over 97% accuracy simply by predicting “no” for every label. The asymmetric loss solves this class imbalance by ignoring these

obvious negatives and forcing the network’s mathematical focus onto the rare, active scent notes that are difficult to predict [19].

3.2.5 Output Phase

The final phase outputs a multi-label binary vector of predictions across all L target scent descriptors which is based on the probability given to each descriptor. A validation module evaluates the model’s performance on these predictions under specified conditions. It follows and calculates metrics such as macro- and micro-averaged F1-scores, Hamming loss, and the Precision-Recall Area Under the Curve (PR-AUC) [3, 48, 51]. These metrics measure the model’s performance, ensuring that it generalizes well to new chemical structures. The three combination methods are compared on the high-confidence (Gold-tier) test set, and also on the full test set for transparency. Each method is trained on all reliability tiers and on Gold-tier blends only, and differences between methods are assessed using paired bootstrap confidence intervals. To test how each method responds to unreliable labels, the models are also retrained on Gold-tier blends with a controlled fraction of labels corrupted, and the decline in macro F1 is compared across methods.

The study additionally reports the results of the label reliability audit: reciprocal agreement rates, cross-catalog agreement by descriptor, and the number of blends and descriptors in each reliability tier.

3.3 Technical Background

Mapping molecular structure onto multi-label olfactory predictions requires stitching together a few different layers of infrastructure: cheminformatics standards to make sure the molecules are valid, a graph learning framework to process them, and a loss function suited to how sparse the labels are. This section walks through each piece—the parsing standards, the sparse graph representation, and the loss objective—and explains why each choice was necessary rather than incidental.

3.3.1 Cheminformatics Standard (RDKit): SMILES Parsing and Sanitization

Molecular structures in this study are represented as Simplified Molecular Input Line Entry System (SMILES) strings, a compact text-based notation, which are canonicalized and parsed into undi-

rected, attributed molecular graphs using the open-source RDKit library in Python [33]. SMILES strings are convenient because they are compact and suitable for storage. This compactness comes at a cost, however: they are structurally sensitive, prone to syntax errors, and do not encode explicit 3D geometry [52]. Before any structure reaches the model, every string passes through RDKit’s sanitization pipeline, which runs four checks in sequence:

- Kekulization checks that the alternating single and double bonds within conjugated or aromatic rings are represented in a chemically consistent manner, resolving ambiguous resonance structures.
- Aromaticity Perception evaluates the coordinate and topological structure of each atom to identify aromatic rings, dynamically reassigning aromatic bond attributes to relevant edges.
- Valence Checking checks every atom, specifically heavy atoms—carbon, nitrogen, oxygen, sulfur, and halogen to ensure it does not exceed its physical valency limits. This rejects syntactically corrupted SMILES strings or physically impossible hypervalent structures.
- Ring Perception, which identifies and indexes ring systems, information the later convolutions depend on.

Enforcing these rules ensures that what the network receives is already known to be chemically valid, rather than requiring the model to infer the rules of organic valency implicitly from training data [4, 59]. The sanitization pipeline acts as an explicit structural filter ahead of the model, so that the Level 1 Graph Attention Network (GAT) operates only on topologically valid graphs that correspond to real molecular structures [27, 55].

3.3.2 Deep Learning Stack (PyTorch / PyTorch Geometric): Sparse Graph Convolutions

After sanitization, each molecule is formatted as an undirected graph $G = (V, E)$, where atoms are nodes (V) and covalent bonds are edges (E). A dense $N \times N$ adjacency matrix could represent this connectivity, but for molecules, where most atoms bond to only a few neighbors, this wastes memory, scaling at $\mathcal{O}(N^2)$ for structures that are inherently sparse. The architecture instead runs on PyTorch Geometric, which handles molecular graphs as sparse attributed objects rather than dense arrays [12].

Topology is stored in Coordinate (COO) format: a sparse two-row index matrix C of size $2 \times 2E$, where the first row lists source node indices and the second lists receiver node indices [4]. Since chemical bonds are physical and undirected, every source node is also a receiver, so $2E$ entries are needed rather than E — each bond is represented in both directions so that message-passing can update both of its atoms. Node states are updated by running the GAT computation described in Section 3.1.2 over this sparse index matrix, rather than over a dense adjacency matrix, at each layer l .

3.3.3 Multi-Label Mathematical Formulation:

Independent Sigmoids and Sparsity Mitigation

A molecule or blend can carry several scent descriptors at once (for example, both fruity and green), rather than belonging to a single exclusive category. To reflect this, the final layer skips a mutually exclusive softmax and instead maps the 256-dimensional blend embedding through an MLP to L independent sigmoid outputs, one per target scent descriptor [39]. Each output is a probability $y_{\text{pred},k}$ bounded between 0 and 1:

$$y_{\text{pred},k} = \sigma(x_k) = \frac{1}{1 + e^{-x_k}}. \quad (3.6)$$

A standard multi-label setup would optimize this with plain Binary Cross-Entropy, summing error across all L target classes independently:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{L} \sum_{k=1}^L \left[y_k \log(y_{\text{pred},k}) + (1 - y_k) \log(1 - y_{\text{pred},k}) \right]. \quad (3.7)$$

Olfactory data is severely imbalanced, however. In the dataset used in this study, label density is approximately 2%, meaning a typical blend has only a few active descriptors out of the full vocabulary, with the remainder absent. Under plain BCE, this large majority of easy negatives dominates the gradient — a model can achieve high overall accuracy simply by predicting zero for every descriptor, and in doing so never learns the rare positive labels that are actually informative.

To address this, the pipeline uses an Asymmetric Focal Loss, which splits the focusing behavior for positives and negatives into separate exponents, γ_p and γ_n , and adds a class-frequency weight α_k [19]:

$$\mathcal{L}_{\text{FL}} = -\frac{1}{L} \sum_{k=1}^L \left[y_k \cdot \alpha_k \cdot (1 - y_{\text{pred},k})^{\gamma_p} \cdot \log(y_{\text{pred},k}) + (1 - y_k) \cdot (1 - \alpha_k) \cdot y_{\text{pred},k}^{\gamma_n} \cdot \log(1 - y_{\text{pred},k}) \right]. \quad (3.8)$$

Setting $\gamma_p = 0$ leaves gradients from the rare positive labels untouched — they pass through at full strength. Setting $\gamma_n = 2$ does the opposite job for negatives: it exponentially suppresses the loss contribution from easy, correctly predicted zeros, stripping away their gradient dominance. Together, these two settings push the model’s training signal toward the rare positive notes instead of allowing it to coast on negatives, which is what makes multi-label prediction feasible on data this sparse [19].

Chapter 4

Methodology

This chapter discusses first discusses how scent is predicted from molecular structure. Next, it tackles how blends of molecules behave and how neural networks handle sets of inputs. Lastly it will describe how the datasets used in this study were built, how reliable their labels are, how evaluation splits are designed, and the gap this study addresses.

4.1 Research Design

This study operates through a Quantitative Experimental Framework, employing supervised learning techniques on a multi-label molecular graph classification task. In this study, the independent variable is the Level 2 aggregation method, with three levels: the Set Transformer (the proposed method), mean pooling, and concatenation following the pairwise design of Sisson et al. [43] (the baselines). A second manipulated variable, used for the third research question, is the training label set (all reliability tiers or Gold only) and the level of injected label noise (0dependent variables: Hamming loss, macro F1, micro F1, and PR-AUC. The controlled variables are the Level 1 GAT encoder, the data split, the loss function, the training hyperparameters, and the random seeds. By doing so, the effect of the aggregation mechanism is isolated in order to ensure that any difference in performance observed could not be attrib The ground truth against which these variables are measured is defined in Section 4.2.3.

4.2 Data Acquisition and Harmonization Pipeline

This section outlines the general procedures for collecting the raw single-molecule data from Leffingwell and GoodScents and blend-level data from Sisson et al.’s [43]. It details how these data are merged, cleaned, and standardized to create one harmonized source.

4.2.1 Dataset Consolidation

Data acquisition will focus on three main datasets: Goodscents, Leffingwell, and Sisson et al.’s [43]. The first two, accessible through PyrFume, serve as the single-molecule sources for Level 1 pretraining and as the lookup reference for the molecular structures referenced in Sisson’s dataset. Sisson et al.’s dataset, accessible through their GitHub repository, serves as the reference for molecule blends used in Level 2 fine-tuning and evaluation.

Each of the two single-molecule datasets contains three files, namely: `behavior.csv`, which contains the stimulus and descriptor pairs data; `molecules.csv`, containing CID, MolecularWeight, IsomericSMILES, IUPACName, name; and the `stimuli.csv` file, which stores data such as Stimulus, TGSC ID, CID, Concentration %, Solvent.

The Leffingwell and Goodscents data files will be merged into one master dictionary, joined by the IsomericSMILES key present in both their `molecules.csv` files, where every molecule has both its canonical structure and its full set of odor descriptors attached. Every IsomericSMILES string in both sources is canonicalized with RDKit so that the same molecule written in different ways is matched correctly. Blend components in Sisson et al.’s dataset, which are identified by PubChem CIDs, are linked to the master dictionary by canonicalizing their SMILES in the same way. This was the key chosen as opposed to CID, as some of the CIDs contained in the files were negative/non-standard and would require further standardization, while the SMILES would already be canonicalized either way in the preparation stage for converting them into graphs. Furthermore, data such as the Concentration % and Solvent from the `stimuli.csv` files are to be dropped, since concentration is scoped out from this study.

Descriptors from `behavior.csv`, which are indexed only by Stimulus, will first be joined to `stimuli.csv` on the shared Stimulus field to obtain each entry’s CID, which is then used to join to `molecules.csv`, which completes the within-source link between a compound’s structure and its descriptors before the cross-source SMILES merge described above takes place.

4.2.2 Vocabulary Standardization

To be more precise, there are approximately 669 unique raw string labels from Goodscents and 114 from Leffingwell, with “unique” here meaning distinct as literal text, since spelling variants of the same underlying descriptor (e.g., anisic/anise) are not yet consolidated at this stage. Upon closer inspection, there is a normalization problem arising wherein sources contain duplicate spellings or variants of scent descriptors, such as anisic/anise, wine/winey, cherry/cherry maraschino cherry, and caramel/caramellic. Manually reviewing each of the scents would be arduous at this scale; therefore, this study will adopt Sisson et al.’s own open-source canonicalization dictionary, which standardizes approximately 400 raw descriptors down to 104.

As the merged data may contain scent descriptors outside this dictionary, a fuzzy string-matching or edit-distance comparison using rapidfuzz’s Levenshtein-based similarity scoring will be done across the residual, uncovered portion of the combined raw label set to identify likely duplicate descriptors. Once identified, they will be manually reviewed and resolved, deciding whether each is mapped to an existing note or declared a genuinely new descriptor, with the latter still subject to being dropped if it does not appear across at least 30 molecules. Further normalization checks will be done, such as whitespace trimming and punctuation stripping.

This standardization applies to the single-molecule (Level 1 pretraining) vocabulary, which is permitted to exceed 104 notes if genuinely new descriptors survive the residual fuzzy-matching process described above. This is distinct from the blend-level (Level 2) classification target, which remains fixed at whatever the resulting count will be when the same survival criterion Sisson et al. applied to their carving run that narrowed down their 104-label vocabulary up to 74 is applied. This paper’s split may yield a similar but not necessarily identical count.

4.2.3 Data Sources, Provenance and Ground Truth

This study uses three data sources. For the single-molecule data, GoodScents and Leffingwell will be used. The third dataset is of two-molecule blends compiled by Sisson et al [43]. from GoodScents. Every molecule is identified by its canonical SMILES string, the ASCII version of its structure, which lets records from different sources refer to the same molecule. GoodScents and Leffingwell & Associates are independently maintained reference catalogs for the perfumery and flavor industries. Each row records one molecule, its SMILES string, and the scent descriptors assigned to it (for

example, fruity or sweet). The descriptors in the Leffingwell PMP 2001 database are described as assigned by olfactory experts, usually practicing perfumers [27]. The two catalogs were merged to train the Principal Odor Map of Lee et al. [27], whose predictions were compared with those of a trained human panel. In this study they train the Level 1 encoder, and the molecules they share are used to measure agreement between the two catalogs.

The blend dataset of Sisson et al. [43] was built by crawling the "Potential Blenders" section of each GoodScents molecule page, which lists the molecules it could be paired with under scent headings. The labels of a pair are the headings under which the partner was listed. The crawl produced more than 160,000 pairs. Each molecule's page was crawled separately. Because of this, a pair can be listed from both molecules' pages. The pair-building code of Sisson et al. [43] counts how many pages listed a pair under each heading (one or two) but does not keep the count. This study keeps the count as a measure of label reliability.

The ground truth in this study relies on scent descriptors from the GoodScents catalog, as compiled by Sisson et al. [43]. Because these are curated reference-catalog records rather than live sensory panel measurements, they serve as a practical proxy for the perceived scent of a blend. To ensure the data's credibility, the reliability of these labels is explicitly measured (Section 4.3) rather than assumed, and the descriptors are grounded by mapping them to a published expert odor taxonomy (Section 3.1.6) [40].

For the computational scope of this study, each blend is treated simply as a physical mixture of two aroma chemicals. We expect no significant chemical reactions between the molecules under standard blending conditions (such as room temperature), meaning the dataset models standard fragrance blending rather than complex chemical reaction processes. While a few highly reactive pairs might deviate from this expectation, this baseline assumption keeps the focus squarely on computational prediction.

The single-molecule labels are drawn from two established industry catalogs: the Leffingwell database, which was compiled by practicing perfumers, and GoodScents. These sources are standard in computational olfaction; previous models trained on them have successfully matched human sensory panels, and recent machine learning benchmarks continue to rely on them. However, because these databases use predefined labels without reporting exactly how human judgments were combined, they inherently contain some data noise and subjectivity [6]. To account for this, we empirically measured their consistency. Across 2,225 shared molecules, the two catalogs agree mod-

erately, showing a mean Jaccard similarity of 0.30, with 64 out of 104 descriptors achieving at least a "fair" kappa score. Recognizing these data limitations, this study treats the catalog labels as a practical proxy for perceived scent, explicitly grades their consistency, and evaluates how the model's predictive performance holds up as label reliability declines

4.2.4 Scent Label List

The model predicts a fixed list of scent labels, detailed in full in Appendix B. This vocabulary comes from the Sisson et al. blend dataset [43], which uses the headings GoodScents assigns to its suggested blending partners. Sisson et al. standardized 109 raw notes down to 104 by merging spelling variants and dropping notes too rare for training and testing [43]. This study applies that same filtering logic to its own data split and reports the final resulting number of labels. For the single-molecule data, the Leffingwell descriptions were standardized into 113 distinct descriptors by Sanchez-Lengeling et al. [27], which are used alongside the raw GoodScents vocabulary. For comprehensive tracking, Appendix B reports how often each blend label appears, whether it is used in the single-molecule catalogs, and its mapped scent family within the Sajan et al. odor taxonomy [40].

4.3 Label Reliability Grading

To ensure data quality, every descriptor on a blend is assigned a reliability tier: Gold, Silver, or Bronze, before any model is trained. This tiering measures consistency (whether independent sources agree) rather than absolute correctness. The grading relies on two strict metrics:

- **Reciprocal Confirmation:** Using the pair-building code of Sisson et al. [43], we check if a blend is listed on both parent molecules' catalog pages. Out of 235,040 total listings, 61,838 (26.3%) were reciprocal (a count of two). While a one-sided listing is not necessarily incorrect, since catalogs do not list every possible blend, it is treated here as not independently confirmed.
- **Cross-Catalog Agreement:** Agreement between GoodScents and Leffingwell was measured using Cohen's kappa across the 2,225 molecules shared between both catalogs. Of the 104 standardized descriptors, 64 achieved a kappa of 0.21 or higher.

4.4 Technical Modeling and Architecture

This section presents the technical architecture of SA-HGNN in the order data flows through it: first Level 1, the Graph Attention Network that encodes each molecule independently from its atomic and bond-level features; then Level 2, the Set Transformer that allows the two resulting molecule embeddings to interact; and finally how the combined blend embedding is converted into per-descriptor predictions, along with the loss function used to train it.

4.4.1 Level 1 GAT Featurization

Once the dataset is read, the merged GoodScents and Leffingwell molecules are first clustered and split using UMAP and K-Means (Section 4.5.2), and only the molecules in the training clusters enter the Level 1 GAT featurization phase for standalone pretraining, so molecules held out for validation and testing are never seen during pretraining. Each molecule is represented as a graph with 46-dimensional node features and 10-dimensional edge features (Section 3.1.1, Appendix A, Tables A.1–A.2). This GAT has three layers and uses LeakyReLU to compute raw attention scores between connected atoms, and normalizes these scores with the softmax formulation in Section 3.1.2.

Once the GAT processes a molecule’s graph, via a mean-pooling readout, its resulting 256-dimensional molecule vector passes through a Multi-Layer Perceptron that maps it to one output per standardized single-molecule descriptor, producing independent sigmoid outputs; the exact number of descriptors is reported after vocabulary standardization (Section 4.4.3).

Afterwards, a Binary Cross-Entropy loss function will be employed for backpropagation. This will be used as opposed to Focal Loss because the latter is more geared toward preventing models from becoming lazy and predicting all-negative outputs, a problem that worsens after Level 2’s graph-carving step. Focal Loss will instead be reserved for Level 2, described in Section 4.3.3.

After pretraining is completed, this temporary output head is discarded, leaving only a GAT with its encoder weights trained. In this step, the model is trained with the single-molecule dataset until it is ready for Level 2, wherein it enters a two-phase protocol for learning blend mechanics.

4.4.2 Level 2 Set Transformer Configuration

With Level 1 delivering a GAT trained on single molecules, the same GAT is applied to the component molecules of a blend pair, once per molecule, producing a 256-dimensional embedding for each:

$\mathbf{z}_A, \mathbf{z}_B \in \mathbb{R}^{256}$ Using the same GAT for both molecules ensures they are embedded under identical rules regardless of order. At this point, the two embeddings form an unordered set:

$$S_{\text{blend}} = \{\mathbf{z}_A, \mathbf{z}_B\} \subset \mathbb{R}^{256} \quad (4.1)$$

and it is the Set Transformer’s subsequent processing, rather than the shared GAT alone that guarantees the blend prediction itself is permutation-invariant: $\{\mathbf{z}_A, \mathbf{z}_B\} = \{\mathbf{z}_B, \mathbf{z}_A\}$ must yield the same result.

This set is passed through the Self-Attention Block (SAB), computing standard attention using Queries (Q), Keys (K), and Values (V) derived from the set itself:

$$\text{SAB}(X) = \text{MultiHead}(X, X, X) \quad (4.2)$$

As in the attention mechanism introduced in Section 3.1.2, each element attends to every other element in the set by comparing its Query against the Keys of all elements (including itself) to compute attention weights, which are then used to combine their Values into an updated representation. Because the same set X serves as the source of Q, K, and V simultaneously, this is self-attention rather than attention across two distinct sets. Concretely, for the two-element blend set, molecule A’s Query is compared against both A’s and B’s Keys, producing attention weights that combine A’s and B’s Values into an updated embedding for A with the same process occurring for B.

SAB is configured with 2 blocks and 4 attention heads, where the two embeddings are no longer treated as individual molecules but as components of a blend. Here, molecules A and B “learn” from each other via message passing, using 4 attention heads in parallel per block, across 2 sequential blocks, with each pass updating both embeddings based on the other’s influence. Formally, this yields the updated embeddings $\mathbf{z}'_A$ and $\mathbf{z}'_B$, where each is now a function of both original embeddings rather than of its own molecule alone. By the end of this process, $\mathbf{z}'_A$ and $\mathbf{z}'_B$ are no longer independent, having been shaped by each other’s influence, designed to allow the model to represent the the masking and emergence described in Chapter 2.

Afterwards, $\{\mathbf{z}'_A, \mathbf{z}'_B\}$ undergoes Pooling by Multihead Attention (PMA), compressing the set into a single 256-dimensional blend embedding $\mathbf{z}_{\text{blend}} \in \mathbb{R}^{256}$. This compression uses attention instead of simple averaging so that it is possible for one molecule’s contribution to be weighted more heavily than the others depending on the learned relationship between them (e.g., a dominant, overpowering scent molecule masking a weaker one).

4.4.3 Output Head & Asymmetric Focal Loss:

The final blend embedding $\mathbf{z}_{\text{blend}} \in \mathbb{R}^{256}$ is passed through a new Multi-Layer Perceptron that contracts from $256 \rightarrow 128 \rightarrow L$, where L is the number of descriptors that survive the leakage-resistant OOD split described in Section 4.5, with up to 74, following Sisson et al.’s own carving outcome, with the exact value reported once this study’s own split is executed.. The output layer applies independent sigmoid activations across all L nodes, $x \in [0, 1]$, since scent descriptors are multi-label: each descriptor must be treated as an independent binary decision ($1 = \text{present}$, $0 = \text{absent}$), unlike softmax, which forces all outputs to sum to 1.0, which is a formulation consistent with the multi-label framework established in Section 3.3.3. The resulting predictions are optimized using the Asymmetric Focal Loss defined in Section 3.3.3, with $\gamma_p = 0$ and $\gamma_n = 2$, following Hatami and Moradi [?].

As such, the final pipeline proceeds as follows for fine-tuning: single-molecule data is featurized and converted into a graph for Level 1; the same GAT trained on single molecules is then reused on those same molecules as components of a blend pair, aggregated together via SAB and PMA, and passed through the L -node sigmoid output head; predictions are scored using Asymmetric Focal Loss, with the resulting gradients updating weights across the entire network, including the pretrained Level 1 GAT end-to-end.

4.4.4 Comparison Baselines

The Set Transformer is compared with two baselines. All three models use the same pretrained Level 1 GAT encoder, the same output head design and the same loss, so only the combining step differs. The first baseline is mean pooling, which calculates the unweighted arithmetic mean of the component vectors, assuming that the perception of a scent mixture is a linear combination of its single-molecule components, using the formula:

$$\mathbf{z}_{\text{blend}} = \frac{1}{M} \sum_{m=1}^M \mathbf{z}_m \quad (4.3)$$

The second baseline uses concatenation, adapting the pairwise design of Sisson et al. [43]. The two Level 1 GAT embeddings are placed side by side to form a 512-dimensional vector, which is then passed through an adapted output head using the same Asymmetric Focal Loss. Because concatenation is inherently sensitive to the order of the inputs, the final predictions are averaged across both possible molecule arrangements.

To strictly isolate the effect of the aggregation method, this baseline is not a direct 1:1 reproduction of Sisson et al.’s model. It intentionally uses our controlled GAT encoder rather than their original encoder, and it is trained and evaluated on our strict UMAP-clustered out-of-distribution (OOD) split rather than their random 50:50 split.

Additionally, sum pooling is excluded from the baseline suite; for strictly two-molecule blends, it is mathematically equivalent to twice the mean and provides no unique predictive information beyond mean pooling.

4.5 Leakage-Resistant OOD Splitting and Graph Carving

This section describes how the dataset is partitioned into training, validation, and test sets such that the resulting evaluation genuinely measures generalization to unseen chemistry, directly addressing the second research question. It explains the reason for why conventional scaffold split is unsuitable for this dataset, then describes the chemical-fingerprint clustering approach used instead, the graph-carving procedure that removes any blend pair spanning two partitions, the descriptor survival filter applied after carving, and finally a sanity check comparing this split against a standard random split.

4.5.1 Scaffold Split Decay

As discussed in more detail in Section 2.6, scaffold splitting does not work well for out-of-distribution (OOD) evaluation in sparse chemical spaces, where scaffold groups contain too few constituent compounds, causing it to collapse to just random splitting instead. This step verifies this claim mathematically as opposed to just relying on the aforementioned literature. To do so, the molecule-to-scaffold ratio (R), is defined as:

$$R = \frac{N_{\text{molecules}}}{N_{\text{scaffolds}}} \quad (4.4)$$

where $N_{\text{molecules}}$ is the total number of unique canonicalized compounds, and $N_{\text{scaffolds}}$ stands for how many distinct Bemis–Murcko frameworks were generated using RDKit. As R gets closer to 1.0, it means that almost every scaffold only contains one molecule, meaning that there is virtually no shared molecular framework across the compounds. R will be computed based on the merged Goodscents + Leffingwell master dictionary during data prep and will be benchmarked against Fooladi et al.’s own collapse threshold of $R < 2.0$. Our R falling below theirs would further justify

using the clustering-based alternative (Morgan-UMAP K-Means graph carving) detailed in the next section, rather than scaffold splitting.

4.5.2 Morgan-UMAP K-Means Split

To represent each molecule, ECFP4 fingerprinting will be performed via RDKit for all unique molecules in the dictionary, generating 2048 binary flags with a radius of 2, capturing atomic environments up to two bonds away.

These 2048-bit ECFP4 fingerprint vectors will then be mapped into a continuous 2D coordinate space using Uniform Manifold Approximation and Projection (UMAP), making it easier to cluster, with chemically and structurally similar molecules sitting closer together and vice versa. K-Means clustering will then be applied to the 2D map with $k = 7$. This value of $k = 7$ follows Guo et al. [16], who found seven UMAP clusters to outperform alternative clustering methods on chemical molecule data.

The OOD test-cluster selection rule works as follows: among the 7 candidate clusters, only those satisfying the requirement that every retained descriptor has at least one training instance and one test instance after graph-carving are considered eligible. Among these eligible candidates, the cluster selected is whichever one retains the largest number of usable blend pairs, as opposed to selecting based on raw cluster size. This mirrors Sisson et al.’s own justification for their chosen split ratio, which similarly prioritized maximizing usable data and label coverage over a fixed proportion.

A second eligible cluster, the one retaining the next-largest number of usable blend pairs, is withheld as the validation cluster, and the remaining clusters form the training set in the selected test and validation clusters are withheld from training, establishing the single-molecule boundary required for the Graph-Carving procedure described in Section 4.4.3.

4.5.3 The Graph-Carving Protocol

The Sisson et al. [43] pairwise blend dataset is treated as a meta-graph in the construction of the graph-carving procedure, wherein each node represents a unique canonical single molecule, and each edge represents a documented blend between two such molecules.

Carving is then performed by examining each edge: if the two molecules an edge connects do not lie within the same partition (training, validation, or test), that edge is discarded. Edges are retained only if both molecules of a blend pair belong to the same partition to ensure there is no

component overlap between any two partitions, preventing leakage. The validation set will be used for early stopping and will be the basis for hyperparameter tuning and the decision threshold for the sigmoid outputs. The test set will be used only once, in the final evaluation.

This carving mechanism mirrors the approach Sisson et al. [43] applied to their own random 50:50 partition, which discarded approximately 83,000 pairs to yield 44,000 training and 40,000 testing pairs. Nevertheless, since this thesis will group its chemical space using the Morgan-UMAP K-means clustering method mentioned in section above, instead of a randomly split 50:50 set, the exact pair count and the label count that will survive the pipeline will only be noted after it is executed.

4.5.4 Survival Vocabulary Filter

With graph carving discarding a significant amount of data from the dataset, the frequencies at which descriptors appear are altered. There is a need for a filter strictly targeting the blend-level target matrix, ensuring that every descriptor in the final multi-label target vector is sufficiently represented in its positive appearances on both sides of the carved split, for meaningful training and evaluation. This is determined by:

$$d \in \mathcal{D}_{\text{eval}} \iff c_{\text{train}}(d) \geq x \quad \text{and} \quad c_{\text{test}}(d) \geq x \quad (4.5)$$

wherein $c_{\text{train}}(d)$ is the number of instances descriptor d is positive in the carved training partition $\mathcal{E}_{\text{train}}$, and $c_{\text{test}}(d)$ is the number of instances descriptor d is positive in the test partition. x , where x , set to $x = 5$, meaning each retained descriptor must be positive in at least 5 blends on both the training and test side. x is the minimum support threshold, where x dictates the minimum number of positive blends required on each side. A descriptor failing this condition on either side $c_{\text{train}}(d) < x$ or $c_{\text{test}}(d) < x$ is dropped from the classification target matrix. This emulates the filtering logic Sisson et al. [43] applied to their own carving run, where their final set of 74 emerged from applying this same survival criterion to their standardized 104-label vocabulary.

4.6 Experiments

To ensure any performance differences are strictly due to the aggregation method, all experiments hold the Level 1 GAT encoder, data splits, training setup, metric suite, and statistical protocol

constant. These tests will determine if explicit attention based interaction modeling statistically outperforms linear pooling and concatenation, or if simple averaging is sufficient for this task.

4.6.1 Experiments A and B: Main Comparison

Every model relies on the same pretrained graph attention encoder to map each molecule to a 256 dimensional embedding. They also share an identical output layer trained with an asymmetric focal loss. The models differ strictly in how they combine the two molecule embeddings:

- Mean pooling: Averages the two embeddings.
- Concatenation: Places the embeddings side by side following the pairwise design of Sisson et al. [43]. Because concatenation is sensitive to input order, final predictions are averaged across both possible molecule arrangements.
- Set Transformer: Uses a self-attention block allowing each embedding to adjust the other, followed by pooling by multihead attention to combine them using learned weights.

In Experiment A, models are trained on all reliability tiers. In Experiment B, they are trained exclusively on Gold descriptors. All configurations are tested on the held out cluster and run across three random seeds.

4.6.2 Experiment C: Label Noise

To evaluate robustness against label noise, a specific fraction ($p = 0, 0.10, 0.20, 0.30$) of the positive labels in the Gold training data is flipped to absent. An equal number of randomly chosen absent labels is flipped to present to keep the overall label density constant. Each model is retrained at each noise level across three random seeds and evaluated on the unmodified test data. The performance drop in macro F1 from $p = 0$ to $p = 0.30$ is then compared across models using paired bootstrap confidence intervals.

4.6.3 Blend-Type Analysis

To evaluate how performance varies by interaction type, the descriptors of each test blend are compared against the independent descriptors of its two parent molecules.

- **Suppression:** A descriptor from either parent is missing in the blend.
- **Emergence:** The blend exhibits a new descriptor not present in either parent.
- **Additive:** The blend shows neither suppression nor emergence.

The relative sizes of these three groups are reported, and the model comparison is repeated within each specific group. If any group contains fewer than 5% or more than 95% of the test blends, a fallback method is used where blends are instead divided based on the overlap between the blend descriptors and the union of the parents’ descriptors.

4.7 Quantitative Metrics and Metric Invariance

This section defines the metrics used to evaluate model performance and explains why each is necessary given the severe label sparsity of the blend-prediction task. It first examines the limitations of precision, recall, and F1 under this sparsity and motivates the use of macro-averaged F1 as the primary metric, then introduces Hamming loss as a complementary metric and defines it formally.

4.7.1 Multi-Label Metric Invariances

Let TP (true positive) be a descriptor correctly predicted as present, FP (false positive) a descriptor incorrectly predicted as present, FN (false negative) a present descriptor the model missed, and TN (true negative) a descriptor correctly predicted as absent. Because only about 2 to 4 of the L descriptors are positive in a typical blend, an estimated 95% of all label entries are negative. Because of this, a model can look like its accurate simply by saying a label is “absent” every time.. Precision, recall and F_1 are computed only from TP, FP and FN [36], so they ignore the many easy true negatives. A model that predicts no descriptors has no true positives and therefore receives an F_1 of zero, which exposes this failure. For this reason, macro-averaged F_1 , which gives every descriptor equal weight regardless of how common it is [36], serves as the primary metric. Micro-averaged F_1 pools all predictions and summarizes overall performance, and PR-AUC measures how well the model ranks true descriptors above false ones across all decision thresholds.

4.7.2 Hamming Loss Integration

Hamming loss is also reported, as it is a standard multi-label metric [36] that counts every incorrect label decision over the full $N \times L$ label matrix. Because it also counts the many easy true negatives, a model that predicts no descriptors would still obtain a low Hamming loss of about 0.03. Hamming loss is therefore interpreted alongside the all-negative and label-frequency reference baselines rather than on its own. This is calculated by the formula:

$$\text{HL} = \frac{1}{N \cdot L} \sum_{i=1}^N \sum_{j=1}^L \mathbf{1}(y_{ij} \neq \hat{y}_{ij}) \quad (4.6)$$

Where N is the total number of evaluated blend samples, L is the number of surviving target descriptors, up to 74 per Sisson et al.’s own carving outcome (Section 4.3.3), with the exact value reported once this study’s own carving is executed $\hat{y}_{ij} \in \{0, 1\}$ is the model’s binary prediction given a decision threshold, and $\mathbf{1}(\cdot)$ is the indicator function (returning 1 if true, 0 if false). Smaller values indicate higher accuracy ($\text{HL} \in [0, 1]$, with $\text{HL} = 0$ representing a perfect match).

4.7.3 Scoring on Gold Labels

Models are scored on the Gold descriptors of the test blends. Each descriptor that is absent from a test blend counts as a negative. Silver and Bronze descriptors of a test blend are ignored, so a model is neither rewarded nor penalized for predicting them. The number of test pairs and descriptors that remain is reported, and scores on all test descriptors are reported alongside for transparency. Macro F1 is the primary metric; micro F1, PR-AUC and Hamming loss are also reported.

4.8 Statistical Significance Testing

This section addresses the risk that the metrics gathered from evaluation might reflect random noise, such as seeding or splitting luck, rather than real performance differences, using inferential statistical testing. Recent work cautions against relying on naive paired t -tests for comparing machine learning models, since the t -test is strongly affected by outliers and is not valid when resampled test sets are used [36]. Instead, this study will employ bootstrap resampling, as recommended by recent work, since it shows both whether one model is better than another and by how much [5]. The Set Transformer and the baseline models will be scored on the same resampled test indices, so that any

difference reflects the models themselves rather than the test samples drawn. To account for seeding luck, each model will also be trained with several random seeds.

To do so, the fixed, held-out test predictions of N blend instances are taken, and N predictions are sampled with replacement 10,000 times, wherein both the Set Transformer and the baseline model are evaluated on the same resampled indices to ensure a paired comparison. For each bootstrap iteration b , the paired metric difference $\Delta(b)$ is computed by:

$$\Delta(b) = \text{Metric}_{\text{SetTransformer}}(b) - \text{Metric}_{\text{Baseline}}(b) \quad (4.7)$$

yielding a distribution of 10,000 paired differences that characterizes the variability of the Set Transformer’s advantage over the baseline. This procedure is repeated independently for each metric in the 4-metric suite from Section 4.5.2.

The significance threshold will be set at $\alpha = 0.05$ and 95% Empirical Confidence Intervals (CIs), which show the plausible range of the true underlying performance difference, will be extracted from the 2.5th and 97.5th percentiles of the 10,000 bootstrap iterations. Here, a CI that does not cross zero indicates a statistically significant positive gain at $\alpha = 0.05$. Since statistical significance alone does not show whether a difference is practically meaningful, the size of the effect is reported as the mean difference together with its 95% CI, expressed in the units of each metric.

This significance testing framework will be deployed for the purpose of comparing the Level 2 Set Transformer against both baselines (mean pooling and concatenation) across the four-metric suite

Appendix A

Chemical Feature Vectors

The exact composition of the 46-dimensional node feature vector and the 10-dimensional edge feature vector used for the intra-molecular GAT encoder are detailed below.

Chemical Property	Dimensions	Mathematical / One-Hot Representation & Details
Atomic Identity	9	One-hot encoding of key organic and halogen elements in flavor/fragrance compounds: [C, N, O, S, F, Cl, Br, I, Other].
Formal Charge	1	Scalar representation of the localized net electrical charge on the atom.
Atomic Degree	6	One-hot representation of the number of connected covalent bonds to heavy atoms: [0, 1, 2, 3, 4, 5+].
Explicit Valence	7	One-hot encoding of the total valence electrons committed to bonding in the molecule: [0, 1, 2, 3, 4, 5, 6+].
Hybridization State	6	One-hot representation of the orbital configuration: [sp , sp^2 , sp^3 , sp^3d , sp^3d^2 , Other].
Aromaticity	1	Binary flag indicating if the atom is part of a resonance-stabilized aromatic ring structure.
Attached Hydrogens	5	One-hot encoding of the number of hydrogens bound to this heavy atom: [0, 1, 2, 3, 4+].
Ring Size Participation	6	Binary flags representing specific ring sizes to map cyclic functional groups: [3-ring, 4-ring, 5-ring, 6-ring, 7-ring, 8-ring].
Ring Membership	1	Binary flag indicating whether the atom is enclosed within any ring structure.
Chirality Tag	4	One-hot representation of tetrahedral stereocenter configuration: [Unspecified, Clockwise (CW), Counterclockwise (CCW), Other].

Table A.1: Specification of the 46-Dimensional Atom (Node) Feature Vector (X_i).

Chemical Property	Dimensions	Mathematical / One-Hot Representation & Details
Bond Type	4	One-hot representation of the bond’s electronic sharing structure: [Single, Double, Triple, Aromatic].
Conjugation	1	Binary flag indicating if the bond belongs to a continuous, overlapping pi-electron system.
Ring Membership	1	Binary flag indicating if the bond is closed within a ring structure.
Stereochemical Config	4	One-hot representation of spatial configuration around double bonds: [None, Stereo-Z, Stereo-E, Other].

Table A.2: Specification of the 10-Dimensional Bond (Edge) Feature Vector (E_{ij}).

A.1 Symmetry Collapse: Representation-Theoretic Derivation

This appendix provides the full representation-theoretic argument for the expressivity limitation of linear, permutation-invariant readouts at the mixture level, summarised in Section 2 of the main text.

Consider a molecular graph G whose set of node embeddings M occupies a d -dimensional representation space $\mathcal{U}$. The structural symmetries of the graph are captured by its automorphism group $\mathcal{H}$, which collects all permutations of the node indices that leave the graph structure unchanged. When a linear, $\mathcal{H}$ -invariant readout function f is applied to aggregate these embeddings, representation theory requires that it factor through the group-averaging Reynolds operator p_{avg} , defined as [47]:

$$p_{\text{avg}} = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \rho(h). \quad (\text{A.1})$$

The Reynolds operator projects the full representation space $\mathcal{U}$ onto its trivial, symmetry-fixed subspace $\mathcal{U}_1$. As a consequence, any linear $\mathcal{H}$ -invariant map decomposes into a downstream linear map composed with this group-averaging projection [47]:

$$f = f' \circ p_{\text{avg}}. \quad (\text{A.2})$$

This factorization establishes that a linear readout can only operate on the information that survives in $\mathcal{U}_1$. The non-trivial, symmetry-sensitive components $\mathcal{U}_2 \oplus \dots \oplus \mathcal{U}_m$, which encode the node-to-node variation responsible for structural differences between molecules, are irreversibly discarded by the projection [47]. As a result, even when the Level 1 GAT encoder produces embeddings that are genuinely sensitive to structural differences between molecules, that information does not survive a linear mixture-level readout. Two distinct molecular combinations that share the same element-wise sum or mean are mapped to identical blend vectors and become indistinguishable to any downstream classifier, regardless of how expressive the encoder is. This expressivity ceiling motivates replacing linear pooling at the mixture level with an attention-based readout that operates on the full representation space [43, 47].

REFERENCES

- [1] R. ACHEBOUCHE, A. TROMELIN, K. AUDOUZE, AND O. TABOUREAU, *Application of artificial intelligence to decode the relationships between smell, olfactory receptors and small molecules*, Scientific Reports, 12 (2022), p. 18817.
- [2] R. L. ALMEIDA, V. G. MALTAROLLO, AND F. G. F. COELHO, *Overcoming class imbalance in drug discovery problems: Graph neural networks and balancing approaches*, Journal of Molecular Graphics and Modelling, 126 (2024), p. 108627.
- [3] D. AMETA, S. KUMAR, R. MISHRA, L. BEHERA, A. CHAKRABORTY, AND T. SANDHAN, *Odor classification: Exploring feature performance and imbalanced data learning techniques*, PLOS ONE, 20 (2025), p. e0322514.
- [4] A. R. N. AOUICHAOU, F. FAN, S. S. MANSOURI, J. ABILDSKOV, AND G. SIN, *Combining group-contribution concept and graph neural networks toward interpretable molecular property models*, Journal of Chemical Information and Modeling, 63 (2023), pp. 725–744.
- [5] Y. BESTGEN, *Please, don't forget the difference and the confidence interval when seeking for the state-of-the-art status*, in Proceedings of the Thirteenth Language Resources and Evaluation Conference, N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, J. Odijk, and S. Piperidis, eds., Marseille, France, June 2022, European Language Resources Association, pp. 5956–5962.
- [6] A. L. BIERLING, A. CROY, T. JESGARZEWSKY, M. ROMMEL, G. CUNIBERTI, T. HUMMEL, AND I. CROY, *A dataset of laymen olfactory perception for 74 mono-molecular odors*, Scientific Data, 12 (2025), p. 347.
- [7] J. B. CASTRO, T. J. GOULD, R. PELLEGRINO, Z. LIANG, L. A. COLEMAN, F. PATEL, D. S. WALLACE, T. BHATNAGAR, J. D. MAINLAND, AND R. C. GERKIN, *Pyrfume: A window to the world's olfactory data*, bioRxiv, (2022).
- [8] C. CHEN, M. SHENG, H. YUAN, X. PAN, H. TIAN, AND X. LOU, *Perceptual interaction of aroma compounds and their effects on the enhancement of flavor quality in foods: a review*, Trends in Food Science & Technology, 165 (2025), p. 105315.
- [9] S. CHITHRANANDA, J. AMORES, AND K. K. YANG, *Mapping the combinatorial coding between olfactory receptors and perception with deep learning*, bioRxiv, (2026).
- [10] G. COUREAUD, T. THOMAS-DANGUIN, J.-C. SANDOZ, AND D. A. WILSON, *Biological constraints on configural odour mixture perception*, J Exp Biol, 225 (2022).

- [11] R. DATTA, J. NISTANE, A. SOSE, H. SAHU, AND R. RAMPRASAD, *Machine learning for green solvents: Assessment, selection and substitution*, Advanced Science, (2025).
- [12] M. FEY AND J. E. LENSSEN, *Fast graph representation learning with pytorch geometric*, CoRR, abs/1903.02428 (2019).
- [13] H. FOOLADI, T. N. L. VU, M. MATHEA, AND J. KIRCHMAIR, *Evaluating machine learning models for molecular property prediction: Performance and robustness on out-of-distribution data*, Journal of Chemical Information and Modeling, 65 (2025), pp. 9871–9891.
- [14] Y. FUKUTANI, M. ABE, H. SAITO, R. EGUCHI, T. TAZAWA, C. A. DE MARCH, M. YOHDA, AND H. MATSUNAMI, *Antagonistic interactions between odorants alter human odor perception*, Current Biology, 33 (2023), pp. 2235–2245.e4. doi: 10.1016/j.cub.2023.04.072.
- [15] D. GASSER, S. SINGH, J. FREIHERR, AND A. GRASSKAMP, *Owsum: algorithmic odor prediction and insight into structure-odor relationships*, Journal of Cheminformatics, 15 (2023).
- [16] Q. GUO, S. HERNANDEZ-HERNANDEZ, AND P. J. BALLESTER, *Umap-based clustering split for rigorous evaluation of ai models for virtual screening on cancer cell lines**, Journal of Cheminformatics, 17 (2025), p. 94.
- [17] R. HAGAN, D. MOORE, F. MCGLONE, AND S. C. WALKER, *Olfactory scene analysis: Does analytical visual processing predict superior identification of component odours in a complex mixture?*, Perception, 54 (2025), pp. 408–430. PMID: 40165414.
- [18] S. HAN, H. FU, Y. WU, G. ZHAO, Z. SONG, F. HUANG, Z. ZHANG, S. LIU, AND W. ZHANG, *Himgnn: a novel hierarchical molecular graph representation learning framework for property prediction*, Briefings in Bioinformatics, 24 (2023), p. bbad305.
- [19] F. HATAMI AND M. MORADI, *Asymmetric focal loss improves graph neural network prediction of drug-drug interactions*, 2026.
- [20] Y. P. HENG, H. Y. LEE, J. W. CHONG, R. R. TAN, K. B. AVISO, AND N. G. CHEM-MANGATTUVALAPPIL, *Incorporating machine learning in computer-aided molecular design for fragrance molecules*, Processes, 10 (2022).
- [21] M. HLADIŠ, M. LALIS, S. FIORUCCI, AND J. TOPIN, *Matching receptor to odorant with protein language and graph neural networks*, in The Eleventh International Conference on Learning Representations, 2023.
- [22] B. HU, H. ZHENG, H. LIANG, H. LV, Q. XIAO, Y. JIANG, Q. LU, Y. NIU, H. WANG, AND Z. XIAO, *Elucidating odorant masking effects in food matrices through olfactory receptor-based profiling*, bioRxiv, (2026).
- [23] J. HUANG, J. LIN, R. YUENG, S. WU, L. SOLLA, AND T. ACREE, *Masking effects on isovaleric acid recognition by sub-threshold odor mixture*, Chemical Senses, 48 (2023), p. bjad047.
- [24] R. JOERES, D. B. BLUMENTHAL, AND O. V. KALININA, *Data splitting to avoid information leakage with datasail*, Nature Communications, 16 (2025), p. 3337.

- [25] A. KENGKANNA AND M. OHUE, *Enhancing property and activity prediction and interpretation using multiple molecular graph representations with mmgx*, Communications Chemistry, 7 (2024), p. 74.
- [26] M. LALIS, M. HLADIŠ, S. A. KHALIL, L. BRIAND, S. FIORUCCI, AND J. TOPIN, *M2or: a database of olfactory receptor-odorant pairs for understanding the molecular mechanisms of olfaction*, Nucleic Acids Research, 52 (2024), pp. D1370–D1379.
- [27] B. K. LEE, E. J. MAYHEW, B. SANCHEZ-LENGELING, J. N. WEI, W. W. QIAN, K. A. LITTLE, M. ANDRES, B. B. NGUYEN, T. MOLOY, J. YASONIK, J. K. PARKER, R. C. GERKIN, J. D. MAINLAND, AND A. B. WILTSCHKO, *A principal odor map unifies diverse tasks in olfactory perception*, Science, 381 (2023), pp. 999–1006.
- [28] J. LEE, Y. LEE, J. KIM, A. R. KOSIOREK, S. CHOI, AND Y. W. TEH, *Set transformer: A framework for attention-based permutation-invariant neural networks*, 2019.
- [29] Q. LI, S. LI, J. LUO, AND H. YUAN, *Predictive models of odor contribution and thresholds for volatiles in identification of novel crop aroma compounds*, Metabolites, 15 (2025).
- [30] K. LÓPEZ-PÉREZ, J. F. AVELLANEDA-TAMAYO, L. CHEN, E. LÓPEZ-LÓPEZ, K. E. JUÁREZ-MERCADO, J. L. MEDINA-FRANCO, AND R. A. MIRANDA-QUINTANA, *Molecular similarity: Theory, applications, and perspectives*, Artificial Intelligence Chemistry, 2 (2024), p. 100077.
- [31] E. MAKRI, N. NAKIS, L. SISSON, G. MINSKY, L. TASSIULAS, V. SATARIFARD, AND N. A. CHRISTAKIS, *Benchmark for assessing olfactory perception of large language models*, 2026.
- [32] C. PATINHA CALDEIRA, R. FARCAL, C. MORETTI, L. MANCINI, H. RAUSCHER, K. RASMUSSEN, J. RIEGO SINTES, AND S. SALA, *Safe and sustainable by design chemicals and materials: Review of safety and sustainability dimensions, aspects, methods, indicators, and tools*, EUR 30991 EN, Joint Research Centre (JRC), European Commission, Luxembourg, 2022.
- [33] A. PAUL, *Dextrosinistral reading of smiles notation: Investigation into origin of non-sense code from string manipulations*, Digital Chemical Engineering, 15 (2025), p. 100222.
- [34] R. PELLEGRINO, E. J. MAYHEW, J. MARGOLIS, M. ANDRES, A. B. WILTSCHKO, R. C. GERKIN, AND J. D. MAINLAND, *Odors smell like their components: A linear framework for predicting olfactory mixture perception*, bioRxiv, (2026).
- [35] S. QIN, S. JIANG, J. LI, P. BALAPRAKASH, R. C. VAN LEHN, AND V. M. ZAVALA, *Capturing molecular interactions in graph neural networks: a case study in multi-component phase equilibrium*, Digital Discovery, 2 (2023), pp. 138–151.
- [36] O. RAINIO, J. TEUHO, AND R. KLÉN, *Evaluation metrics and statistical tests for machine learning*, Scientific Reports, 14 (2024), p. 6086.
- [37] B. C. L. RODRIGUES, V. V. SANTANA, S. MURINS, AND I. B. R. NOGUEIRA, *Molecule generation and optimization for efficient fragrance creation*, Industrial & Engineering Chemistry Research, 63 (2024), pp. 14480–14494.
- [38] S. ROMAGNY, G. COUREAUD, AND T. THOMAS-DANGUIN, *Developmental switch from elemental to configural processing of odor mixtures in humans*, Developmental Psychobiology, 68 (2026), p. e70188.

- [39] K. SAINI AND V. RAMANATHAN, *Predicting odor from molecular structure: a multi-label classification approach*, Scientific Reports, 12 (2022), p. 13863.
- [40] A. SAJAN, S. SLUIS, R. HAYDARLOU, S. ABELN, P. LISENA, R. TRONCY, C. VERBEEK, I. LEEMANS, AND H. MOUHIB, *Exploring molecular odor taxonomies for structure-based odor predictions using machine learning*, 2025.
- [41] E. I. SANCHEZ MEDINA, S. LINKE, M. STOLL, AND K. SUNDMACHER, *Gibbs–helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution*, Digital Discovery, 2 (2023), pp. 781–798.
- [42] H. SHIMAKAWA, A. KUMADA, AND M. SATO, *Extrapolative prediction of small-data molecular property using quantum mechanics-assisted machine learning*, npj Computational Materials, 10 (2024), p. 11.
- [43] L. SISSON, A. A. BARSAINYAN, M. SHARMA, AND R. KUMAR, *Deep learning for odor prediction on aroma-chemical blends*, ACS Omega, 10 (2025), pp. 8980–8992.
- [44] L. J. S. SOBRAL, F. L. FEITOSA, J. A. M. KUNZ, P. K. DINIZ, J. F. AVELLANEDA-TAMAYO, J. L. MEDINA-FRANCO, A. G. FILHO, AND C. H. ANDRADE, *Odor-sight: An interpretable graph neural network web platform for prediction of odorant activity*, Journal of Chemical Information and Modeling, 66 (2026), pp. 9721–9728.
- [45] S. STESHIN, *Lo-hi: Practical ml drug discovery benchmark*, 2023.
- [46] J. SUH, Y. HONG, AND C. PARK, *A comparative study of machine learning models on molecular fingerprints for odor decoding*, Communications Chemistry, 8 (2025), p. 278.
- [47] M. TALHI, A. WOLF, AND A. MONOD, *Breaking symmetry bottlenecks in gnn readouts*, 2026.
- [48] A. N. TAREKEGN, M. GIACOBINI, AND K. MICHALAK, *A review of methods for imbalanced multi-label classification*, Pattern Recognition, 118 (2021), p. 107965.
- [49] G. TOM, C. T. SER, E. RAJAONSON, S. LO, H. S. PARK, B. LEE, AND B. SANCHEZ-LENGELING, *Does this smell the same? learning representations of olfactory mixtures using inductive biases*, Machine Learning: Science and Technology, 6 (2025).
- [50] J. WANG, Q. ZHANG, W. FAN, Q. SHI, J. MAO, J. XIE, G. CHAI, AND C. ZHANG, *Deciphering olfactory receptor binding mechanisms: a structural and dynamic perspective on olfactory receptors*, Frontiers in Molecular Biosciences, Volume 11 - 2024 (2025).
- [51] T. WEN, X. CAI, AND J. LI, *Graph neural networks vs. traditional qsar: A comprehensive comparison for multi-label molecular odor prediction*, Molecules, 30 (2025).
- [52] D. S. WIGH, J. M. GOODMAN, AND A. A. LAPKIN, *A review of molecular representation in the age of machine learning*, WIREs Computational Molecular Science, 12 (2022), p. e1603.
- [53] Y.-X. WU, X. WANG, A. ZHANG, X. HE, AND T.-S. CHUA, *Discovering invariant rationales for graph neural networks*, 2022.
- [54] Z. WU, J. WANG, H. DU, D. JIANG, Y. KANG, D. LI, P. PAN, Y. DENG, D. CAO, C.-Y. HSIEH, AND T. HOU, *Chemistry-intuitive explanation of graph neural networks for molecular property prediction with substructure masking*, Nature Communications, 14 (2023), p. 2585.

- [55] H. XIE, J. SUN, Y. SHAO, S. LI, S. HOU, Y. SUN, AND J. WANG, *Molecular odor prediction based on multi-feature graph attention networks*, 2025.
- [56] L. XU, D.-J. ZOU, AND S. FIRESTEIN, *Odor mixtures: A chord with silent notes*, *Frontiers in Ecology and Evolution*, Volume 11 - 2023 (2023).
- [57] M. ZAHEER, S. KOTTUR, S. RAVANBAKSH, B. POCZOS, R. SALAKHUTDINOV, AND A. SMOLA, *Deep sets*, 2018.
- [58] M. ZHANG, Y. HIKI, A. FUNAHASHI, AND T. J. KOBAYASHI, *A deep position-encoding model for predicting olfactory perception from molecular structures and electrostatics*, *NPJ systems biology and applications*, 10 (2024), p. 76.
- [59] W. ZHU, Y. ZHANG, D. ZHAO, J. XU, AND L. WANG, *Hignn: Hierarchical informative graph neural networks for molecular property prediction equipped with feature-wise attention*, 2022.